MATHEMATICAL BIOLOGY

How Does a Hydra Decide Where to Grow a Head?

Szymon Cygan
Heidelberg university

Chemical and mechanical routes to pattern formation

LECTURE MAP

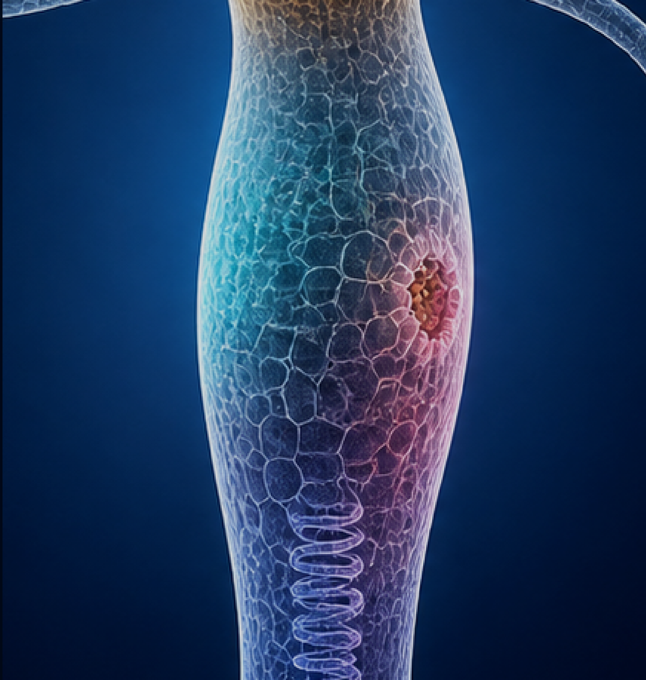
The road from Hydra to pattern formation

From biological observations to mathematical mechanisms.

01 THE BIOLOGICAL PUZZLE

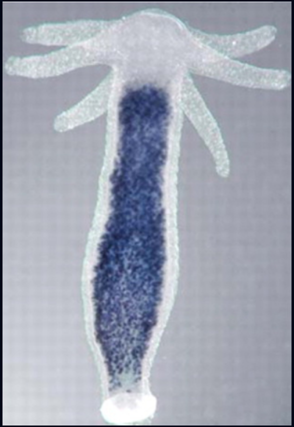
Hydra as a model organism

A simple organism makes symmetry breaking directly visible.



Hydra: simple body, continuous renewal

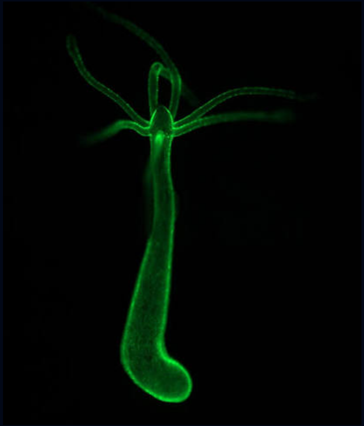
- **A small freshwater cnidarian**
  A transparent, tube-shaped polyp attached to the substrate by a basal disc.
- **Simple oral–aboral anatomy**
  A head with hypostome and tentacles, a body column, and a foot.
- **Two epithelial layers**
  Epidermis outside and gastrodermis inside, separated by mesoglea around one gastric cavity.
- **Continuous tissue renewal**
  Stem cells divide in the body column while cells are displaced toward the head and foot.
- **Evolutionary significance**
  An early-branching animal model for organizers, axial patterning and regeneration.



Augustin R et al., Dev Biol. 2006; 296(1):62-70.

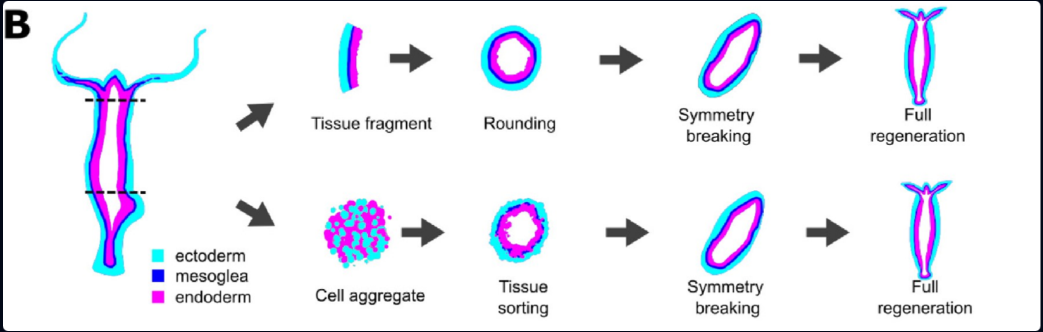
Hydra can rebuild a whole animal from very little

- **Head removal**
  The remaining body column regenerates a new head at the wound.
- **Bisection**
  The apical half rebuilds a foot, the basal half rebuilds a head — producing two complete animals.
- **Small tissue fragment**
  A fragment closes, rounds up, breaks symmetry and reconstructs an oral–aboral axis.
- **Dissociated cells**
  Cells can be separated, reaggregated, sorted into layers and self-organized into new polyps.
- **The central puzzle**
  Where does the new head form when the starting tissue is nearly symmetric?



Wittlieb J et al., Dev Biol. 2006; 296(1):62-70.

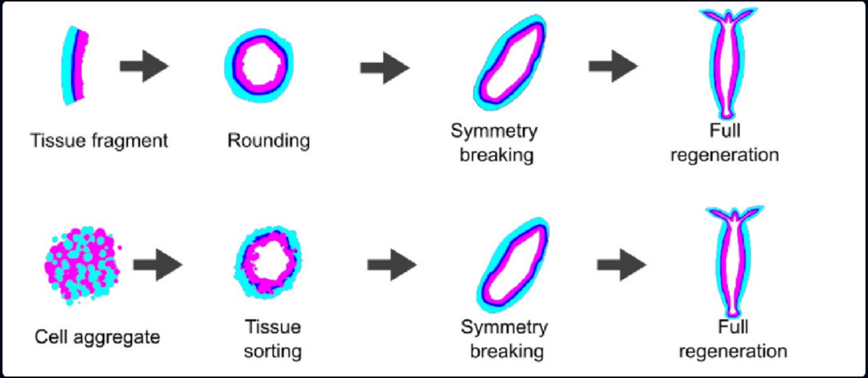
Regeneration unfolds through ordered stages



Wang, Bialas, Goel & Collins, *Integrative and Comparative Biology* 63 (2023), Fig. 1B.

Very different initial conditions pass through a similar spheroid stage before rebuilding a body axis

Symmetry breaks before a new head becomes visible



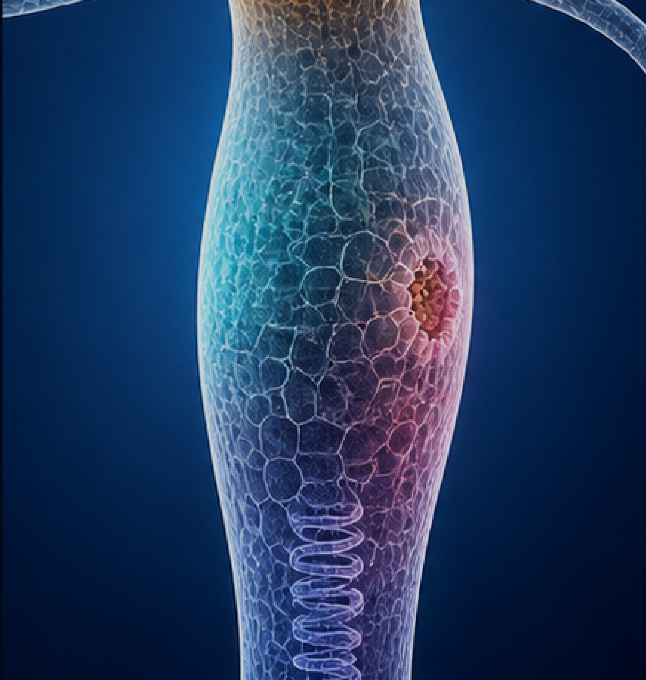
Wang, Bialas, Goel & Collins, *Integrative and Comparative Biology* 63 (2023), Fig. 1B.

A local site acquires head-forming identity first; visible tentacles and the complete head–foot axis appear later

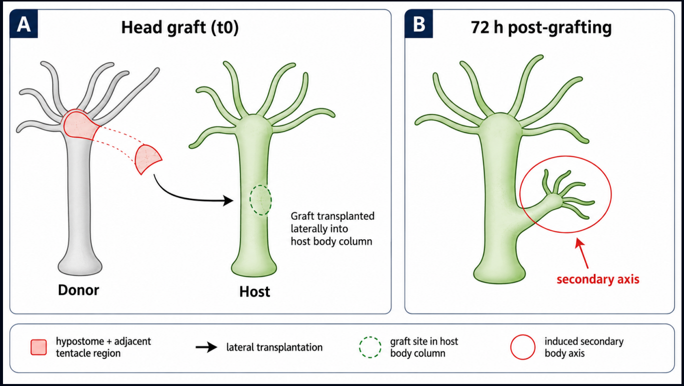
01 THE BIOLOGICAL PUZZLE

Hydra grafting experiments

Transplantation reveals that organizer activity depends on position and context.



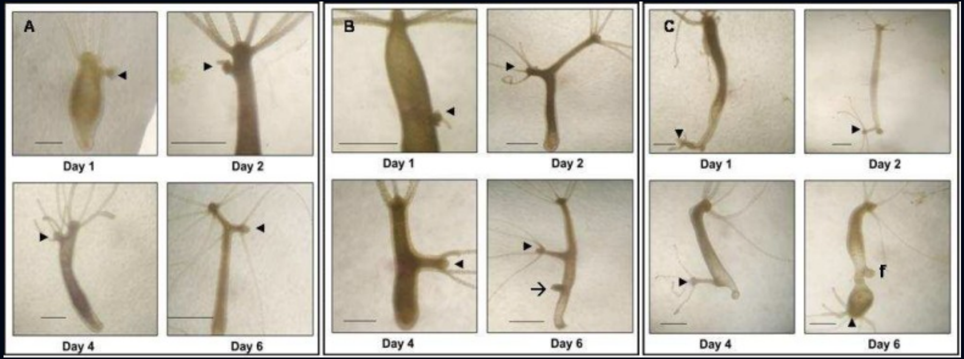
A head graft can organize a second body axis



Schematic based on the classic hypostome transplantation experiment of Browne (1909); see also Broun & Bode (2002).

The graft does not simply grow into a second Hydra; it recruits a new body axis

Impact of the grafting position

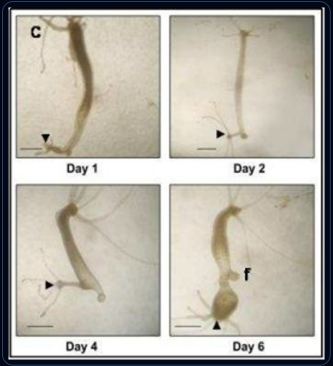


Kadu, Ghaskadbi & Ghaskadbi, *Int. J. Mol. Cell. Med.* 1 (2012).

The same graft can produce different outcomes depending on its position

Far from the host head: a complete second axis

lower host position



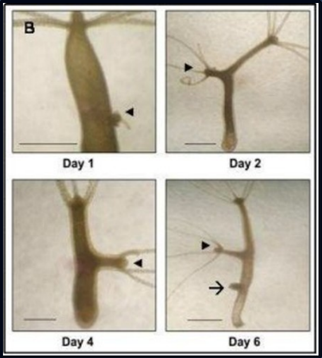
Kadu, Ghaskadbi & Ghaskadbi, *Int. J. Mol. Cell. Med.* 1 (2012), Fig. 2C.

- **A complete secondary axis**
  The induced structure develops a foot, body column, hypostome and tentacles.

- **The strongest response**
  Far from the host head, the same graft produces the longest and most complete secondary axis.

Mid-body graft: an intermediate second axis

middle host position



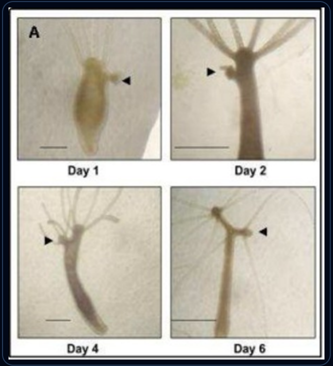
Kadu, Ghaskadbi & Ghaskadbi, *Int. J. Mol. Cell. Med.* 1 (2012), Fig. 2B.

- **An intermediate secondary axis**
  The induced axis grows substantially longer than after an upper-body graft.

- **Only position has changed**
  The donor tissue is the same; moving the graft along the host axis changes the resulting morphology.

Near the host head: only a short second axis

upper host position



Kadu, Ghaskadbi & Ghaskadbi, *Int. J. Mol. Cell. Med.* 1 (2012), Fig. 2A.

- **A very short secondary axis**
  The hypostome graft still induces a new head, but only a short body column develops.

- **Growth is strongly constrained**
  Close to the host head, the induced axis remains small despite using the same organizer tissue.

Multiple grafts can impose multiple axes

- **Three grafts on one host**
  A complete foot was grafted in the upper region, hypostome pieces were grafted in the middle and lower regions.

- **Three secondary axes**
  All three grafted regions produced an induction simultaneously.

- **Organizer identity matters**
  The axes induced by hypostome tissue were substantially longer than the axis induced by the foot graft.

Kadu, Ghaskadbi & Ghaskadbi, *Int. J. Mol. Cell. Med.* 1 (2012), Fig. 6.

Multiple local organizers can coexist and impose several body axes on the same tissue.

From experiments to a mathematical model

regenerating tissue → organizer selection → one body axis

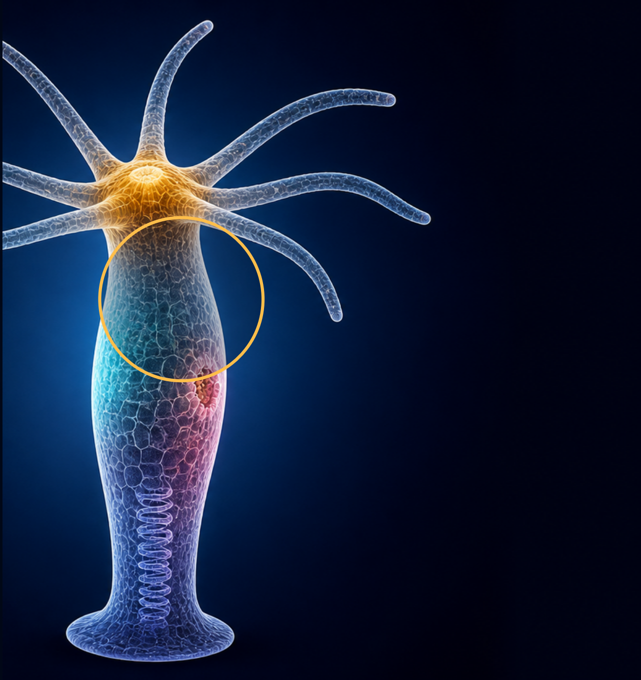
ectopic organizer → interaction with host → secondary axis

- **Spontaneous pattern formation**
  How can a localized organizer emerge from weakly patterned or nearly symmetric tissue?
- **Organizer induction**
  Why can a transplanted organizer recruit host tissue and initiate a new axis?
- **Competition and uniqueness**
  Why does normal regeneration usually select one head, while several grafts can sustain several axes?
- **Context dependence**
  Why does the same hypostome graft produce different secondary axes at different positions in the host?

LECTURE MAP

The road from Hydra to pattern formation

From biological observations to mathematical mechanisms.



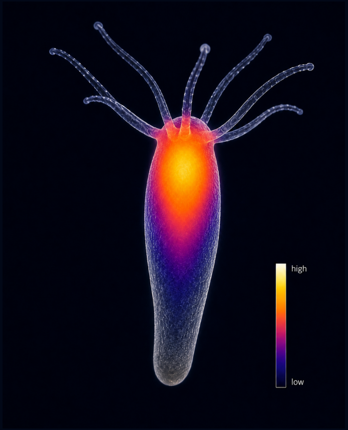
02 REACTION–DIFFUSION APPROACH

Chemical patterning

Chemical signals provide a natural mechanism for spatial organization.

Could chemical signals determine where a head forms?

- **Cells communicate through chemical signals**
  Signaling molecules can be produced, transformed and removed by cells.
- **Concentrations can vary in space**
  Different regions of the tissue may therefore experience different levels of the same signal.
- **A local maximum can select a site**
  A sufficiently high concentration may trigger head-forming activity in one particular region.
- **Chemistry can position the organizer**
  The location of head formation becomes a question of how a spatial concentration pattern arises.



Reactions change signals, diffusion moves them

REACTION

- **Local chemical change**
  Cells can produce, remove or transform signaling molecules.

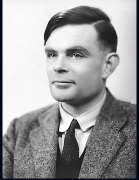
- **Local feedback**
  The production of a signal may depend on the concentrations already present in the same region.

DIFFUSION

- **Spatial redistribution**
  Signaling molecules spread from regions of higher concentration toward regions of lower concentration.

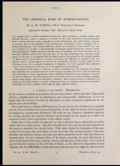
- **Communication across tissue**
  Diffusion couples nearby regions, allowing a local chemical change to influence its surroundings.

Mechanism of chemical pattern formation



1952

Alan Turing



[37]

THE CHEMICAL BASIS OF MORPHOGENESIS

By A. M. TURING, F.R.S. *University of Manchester*

- **A homogeneous tissue need not stay homogeneous**
  Chemical substances can react and diffuse through tissue, allowing spatial structure to emerge from small perturbations.

- **Diffusion becomes part of the mechanism**
  Spatial transport is not merely smoothing: together with reactions it can participate in pattern formation.

- **Hydra was already on the list**
  Turing explicitly mentioned stationary patterns as a possible explanation for tentacle organization in Hydra.

Reaction and diffusion can, in principle, turn an almost homogeneous state into spatial structure

Gierer and Meinhardt propose a model

Alfred Gierer

Hans Meinhardt

A. GIERER & H. MEINHARDT / 1972

A Theory of Biological Pattern Formation

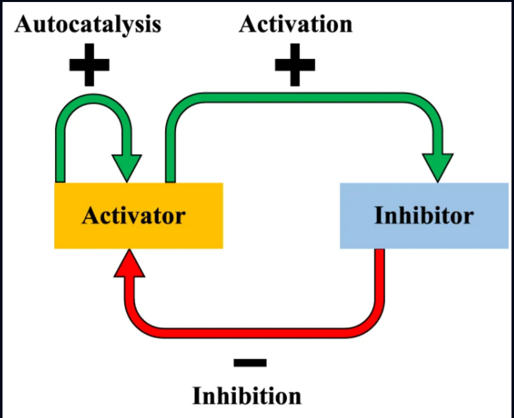
A mechanism for generating spatial organization in an initially almost homogeneous tissue.

- **Local self-activation**
  A small local excess reinforces itself and can develop into a localized organizing region.

- **Long-range inhibition**
  The activated region generates an inhibitory effect that acts over a larger spatial range.

Gierer & Meinhardt, *Kybernetik* 12 (1972), 30–39.

The activator–inhibitor idea



- **Activator**
  Enhances its own production and promotes local organizer formation. Its effect remains relatively localized.
- **Inhibitor**
  Is produced by activated regions, suppresses further activation and acts over a longer spatial range.

The Gierer–Meinhardt model

$$\partial_t u = \underbrace{D_u \Delta u}_{\text{diffusion}} + \underbrace{\frac{au^2}{K+v} - \mu_u u}_{\text{reaction}}$$

$$\partial_t v = \underbrace{D_v \Delta v}_{\text{diffusion}} + \underbrace{bu^2 - \mu_v v}_{\text{reaction}}$$

- ***u*: activator**
  Promotes its own production and initiates local organizer activity
- ***v*: inhibitor**
  Produced by the activator and suppresses its production

D_u, D_v	diffusion rates, typically $D_v > D_u$
a	strength of activator self-amplification
b	strength of inhibitor production
K	scale of inhibition by v
μ_u	activator degradation rate
μ_v	inhibitor degradation rate

A general reaction–diffusion system

$$\partial_t u = \underbrace{D_u \Delta u}_{\text{diffusion}} + \underbrace{f(u,v)}_{\text{reaction}},$$

$$\partial_t v = \underbrace{D_v \Delta v}_{\text{diffusion}} + \underbrace{g(u,v)}_{\text{reaction}}.$$

- ***u* and *v***
  Concentrations of two interacting chemical signals.
- ***f* and *g***
  Possibly nonlinear local production, removal and interaction terms.

We begin by deriving the spatial transport term

02 REACTION–DIFFUSION APPROACH

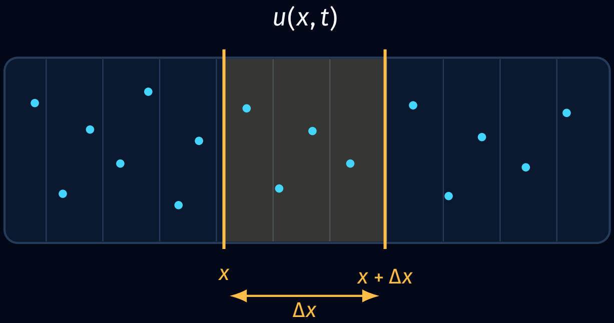
Diffusion

Diffusion redistributes signals through space and smooths concentration differences

Diffusion begins with a local balance

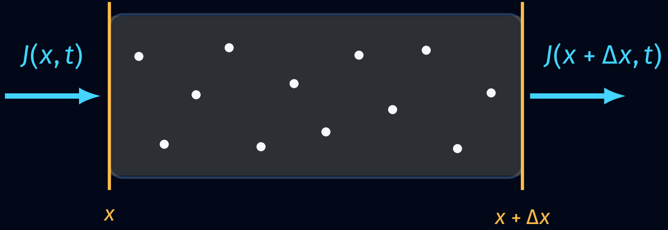
- **Concentration**
  $u(x,t)$ measures the amount of a substance per unit length at point x.

- **Control volume**
  We track the substance inside a short interval of length Δx.



amount in the interval $\approx u(x,t)\,\Delta x$

Boundary flux



Flux J: amount of substance crossing a boundary.

$J(x,t) > J(x+\Delta x,t)$
amount increases

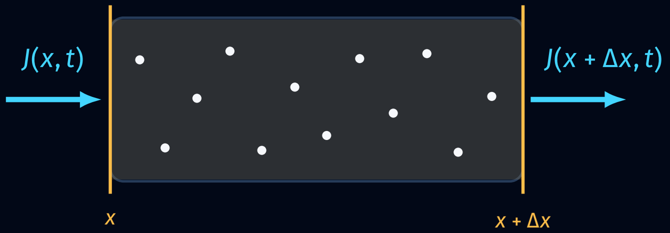
$J(x,t) = J(x+\Delta x,t)$
amount stays constant

$J(x,t) < J(x+\Delta x,t)$
amount decreases

$$\frac{u(x, t+\Delta t)\,\Delta x - u(x,t)\,\Delta x}{\Delta t} = J(x,t) - J(x+\Delta x, t)$$

Only a difference between the boundary fluxes changes the amount inside

The local conservation law



$$\frac{u(x, t + \Delta t) - u(x, t)}{\Delta t} = -\frac{J(x + \Delta x, t) - J(x, t)}{\Delta x}$$

$$\Downarrow \quad \Delta t, \Delta x \longrightarrow 0$$

$$\boxed{\partial_t u = -\partial_x J}$$

Fick's law of diffusion

$$\boxed{J = -D\,\partial_x u}$$

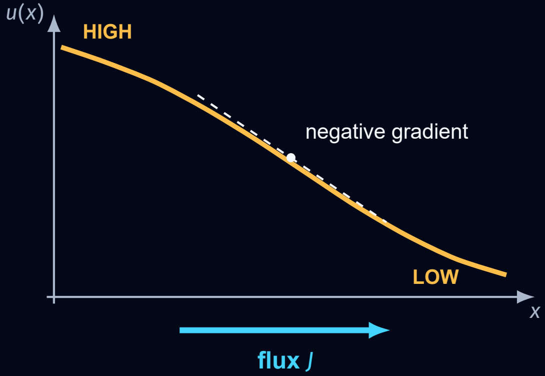
ADOLF FICK

1829–1901

1855

Ueber Diffusion

Using Fourier's heat-conduction analogy, Fick related diffusive flux to the concentration gradient



$\partial_x u < 0 \Rightarrow J > 0$
flux to the right

$\partial_x u = 0 \Rightarrow J = 0$
no concentration-driven flux

$|J| = D\,|\partial_x u|$
steeper gradient, larger flux

Diffusion equation

LOCAL CONSERVATION

$$\partial_t u = -\partial_x J$$

$+$

FICK'S LAW

$$J = -D\partial_x u$$

$\Downarrow$

DIFFUSION EQUATION

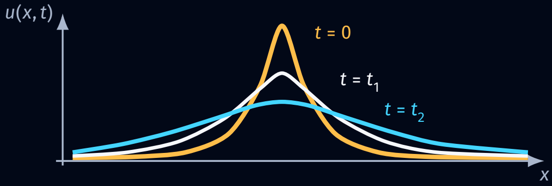
$$\boxed{\partial_t u = D\partial_{xx} u}$$

The diffusion equation on a bounded domain

$\Omega = [0, L] \subset \mathbb{R}$

$$\partial_t u(x,t) = D\,\partial_{xx} u(x,t),$$

$$u(x,0) = u_0(x).$$



- **No-flux boundary (Neumann)**
  The sealed ends prevent the substance from leaving

$$\partial_x u(0,t) = 0 \qquad \partial_x u(L,t) = 0$$

- **Absorbing boundary (Dirichlet)**
  Substance that reaches an open end leaves the domain

$$u(0,t) = 0, \qquad u(L,t) = 0.$$

Fourier turns diffusion into a problem of spatial modes

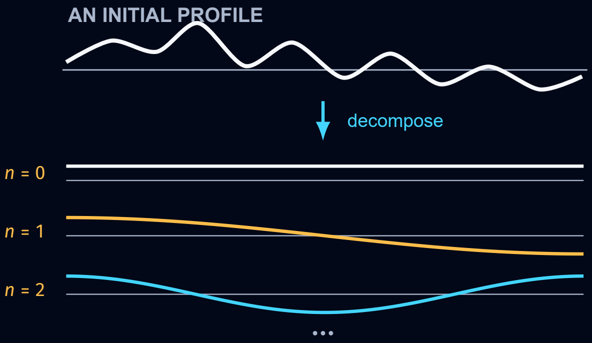
1822

Théorie analytique de la chaleur

- **Fourier's question**
  How does an arbitrary temperature profile evolve under the heat equation?

- **The key idea**
  Decompose a complicated spatial profile into simple sine and cosine waves.

$$u_0(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right).$$



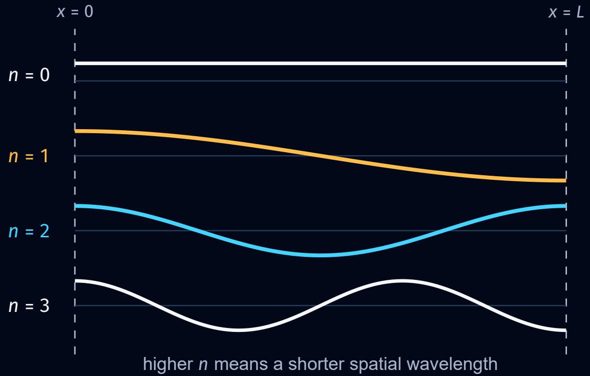
Instead of solving for every initial profile separately, solve how each elementary spatial wave evolves.

No-flux boundaries — cosine modes

- **A mode keeps its spatial shape**
  Applying the diffusion operator changes only its amplitude, not the form of the profile.

$$-\phi_n'' = \mu_n \phi_n,$$
$$\phi_n'(0) = \phi_n'(L) = 0.$$

$$\phi_n(x) = \cos\left(\frac{n\pi x}{L}\right),$$
$$\mu_n = \left(\frac{n\pi}{L}\right)^2.$$



The derivative vanishes at both endpoints, so every cosine satisfies the no-flux boundary conditions.

Eigenfunctions and eigenmodes

$$u(x,t) = \sum_{n=0}^{\infty} a_n(t)\phi_n(x)$$

$$\phi_n(x) = \cos\left(\frac{n\pi x}{L}\right)$$

$$\mu_n = \left(\frac{n\pi}{L}\right)^2$$

$$-\phi_n'' = \mu_n \phi_n \qquad \phi_n'(0) = \phi_n'(L) = 0$$

$$\partial_t u(x,t) = \sum_{n=0}^{\infty} a_n'(t)\phi_n = D\sum_{n=0}^{\infty} a_n(t)\phi_n'' = D\,\partial_{xx}u(x,t)$$

$$\boxed{\begin{aligned} a_n'(t) &= -D\mu_n a_n(t) \\ a_n(t) &= a_n(0)e^{-D\mu_n t} \end{aligned}}$$

- **Short waves disappear first**
  Because $\mu_n = (n\pi/L)^2$, modes with larger n decay much faster.

Diffusion in higher dimensions

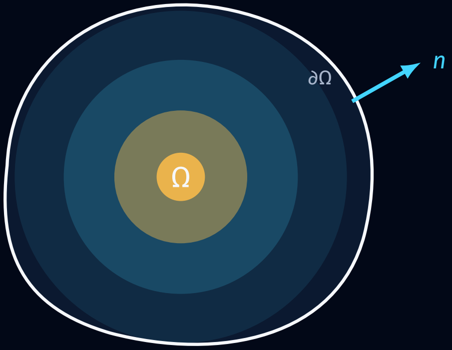
$$x \in \Omega \subset \mathbb{R}^d \qquad \partial_t u = D\Delta u$$

In Cartesian coordinates,

$$\Delta u = \partial_{x_1 x_1} u + \cdots + \partial_{x_d x_d} u$$

- **No-flux boundary**
  A closed tissue does not allow the signal to cross its boundary.

$$\nabla u \cdot n = 0 \qquad \text{on } \partial\Omega$$



substance remains inside the tissue

Eigenfunctions on a bounded domain

THEOREM

There exist eigenfunctions $\{\phi_k\}_{k=0}^{\infty}$ and eigenvalues $\{\mu_k\}_{k=0}^{\infty}$ of the Laplace operator satisfying

$$-\Delta\phi_k = \mu_k\phi_k, \qquad \nabla\phi_k \cdot n = 0 \quad \text{on } \partial\Omega.$$

$$0 = \mu_0 < \mu_1 \leq \mu_2 \leq \cdots, \qquad \boxed{\mu_k \longrightarrow \infty}.$$

The eigenfunctions form an orthonormal basis, solution can be written as a sum of spatial modes.

$$u_0(x) = \sum_{k=0}^{\infty} a_k(0)\phi_k(x),$$

$$u(x,t) = \sum_{k=0}^{\infty} a_k(0)e^{-D\mu_k t}\phi_k(x).$$

- **Different modes, different rates**
  Large eigenvalues correspond to rapidly decaying spatial structure.
- **No mode is amplified**
  Every nonconstant mode decreases under diffusion alone.

Diffusion alone damps spatial variation

02 REACTION–DIFFUSION APPROACH

Local dynamics

Reactions change signals locally through production, degradation and nonlinear interactions

Reaction

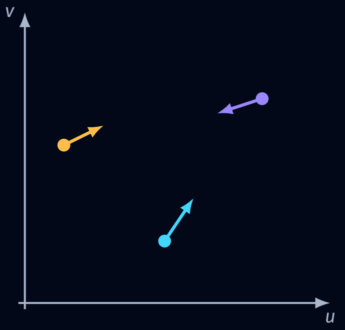
Begin with a system of ordinary differential equations

$$\partial_t u = f(u, v)$$
$$\partial_t v = g(u, v)$$

u and *v* are the local concentrations of two substances

The functions $f(u, v)$ and $g(u, v)$ describe how these concentrations change depending on their current values

- **One state at every position**
  The pair (u, v) represents the local chemical state.



the vector $(f(u, v), g(u, v))$
determines the direction of change

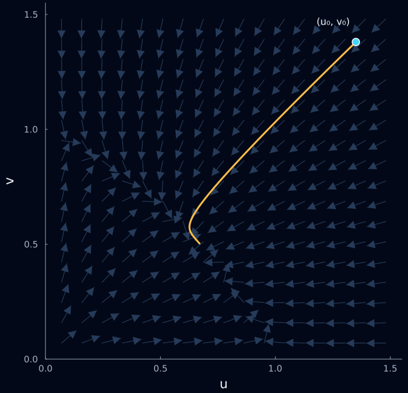
Reaction — vector field and trajectory

$$\partial_t u = f(u,v)$$
$$\partial_t v = g(u,v)$$

supplemented with initial conditions

$$u_0 = u(0) \qquad v_0 = v(0)$$

- **Initial state**
  The pair (u_0, v_0) selects the starting point in the concentration plane.
- **Velocity**
  The vector $(f(u,v), g(u,v))$ gives the instantaneous direction of change.
- **Trajectory**
  The solution curve is tangent to the vector field at every point.



Steady state

At a steady state $u(t) \equiv \bar{u}$, $v(t) \equiv \bar{v}$, both concentrations remain constant in time

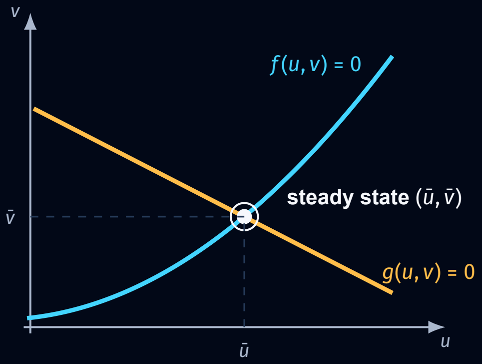
$$f(\bar{u}, \bar{v}) = 0 \qquad g(\bar{u}, \bar{v}) = 0$$

NULLCLINES

$f(u, v) = 0$ — u does not change

$g(u, v) = 0$ — v does not change

Each nullcline contains states for which one concentration is instantaneously stationary



the nullclines organize the local phase plane

Lyapunov stability

Consider two solutions of the same system

$$\partial_t u = f(u,v)$$
$$\partial_t v = g(u,v)$$

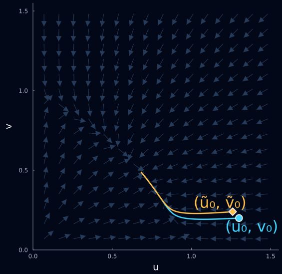
$u(0) = u_0$ $\qquad$ $\tilde{u}(0) = \tilde{u}_0$

$v(0) = v_0$ $\qquad$ $\tilde{v}(0) = \tilde{v}_0$

LYAPUNOV STABILITY

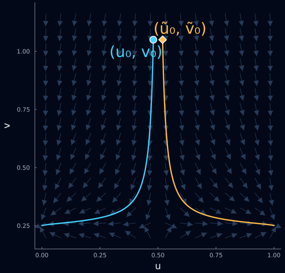
For every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\|(u_0, v_0) - (\tilde{u}_0, \tilde{v}_0)\| < \delta \quad \Rightarrow$$
$$\|(u(t), v(t)) - (\tilde{u}(t), \tilde{v}(t))\| < \varepsilon \qquad \text{for all } t \geq 0$$



STABLE

Nearby solutions converge



UNSTABLE

Nearby solutions separate

Stability of constant steady state

$$u(0) = \bar{u} + \eta_0$$
$$v(0) = \bar{v} + \xi_0$$

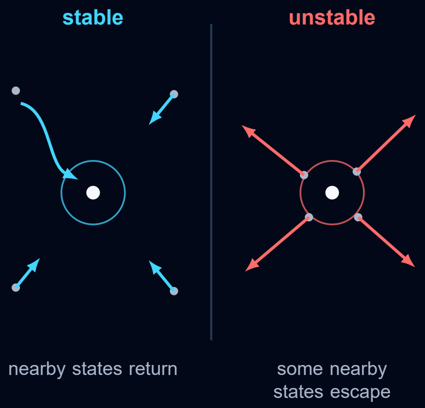
The perturbation is given by

$$\eta(t) = u(t) - \bar{u}$$
$$\xi(t) = v(t) - \bar{v}$$

- **Asymptotically stable**
  Every sufficiently small perturbation returns to the equilibrium.

$$\begin{aligned} \eta(t) &\longrightarrow 0 \\ \xi(t) &\longrightarrow 0 \end{aligned} \quad \text{as } t \longrightarrow \infty$$

- **Unstable**
  At least one arbitrarily small perturbation moves away.



Linearized stability

Near the steady state $(\bar{u}, \bar{v})$ write

$$u = \bar{u} + \eta, \quad v = \bar{v} + \xi$$

TAYLOR EXPANSION

$$\partial_t \eta = f(\bar{u} + \eta, \bar{v} + \xi) = f(\bar{u}, \bar{v}) + f_u(\bar{u}, \bar{v})\eta + f_v(\bar{u}, \bar{v})\xi + O\left(\|(\eta, \xi)\|^2\right)$$
$$\partial_t \xi = g(\bar{u} + \eta, \bar{v} + \xi) = g(\bar{u}, \bar{v}) + g_u(\bar{u}, \bar{v})\eta + g_v(\bar{u}, \bar{v})\xi + O\left(\|(\eta, \xi)\|^2\right)$$

JACOBIAN

$$\boxed{J(\bar{u}, \bar{v}) = \begin{pmatrix} f_u(\bar{u}, \bar{v}) & f_v(\bar{u}, \bar{v}) \\ g_u(\bar{u}, \bar{v}) & g_v(\bar{u}, \bar{v}) \end{pmatrix}}$$

NEGLECTING HIGHER-ORDER TERMS

$$\boxed{\begin{pmatrix} \partial_t \eta \\ \partial_t \xi \end{pmatrix} = J(\bar{u}, \bar{v}) \begin{pmatrix} \eta \\ \xi \end{pmatrix}}$$

Close to the steady state, the Jacobian determines the leading-order dynamics

Eigenvalues of the Jacobian

LINEARIZED SYSTEM

$$\begin{pmatrix} \partial_t \eta \\ \partial_t \xi \end{pmatrix} = J(\bar{u}, \bar{v}) \begin{pmatrix} \eta \\ \xi \end{pmatrix}$$

$$J(\bar{u}, \bar{v}) = \begin{pmatrix} f_u(\bar{u}, \bar{v}) & f_v(\bar{u}, \bar{v}) \\ g_u(\bar{u}, \bar{v}) & g_v(\bar{u}, \bar{v}) \end{pmatrix}$$

The eigenvalues are the roots of

$$\det(J(\bar{u}, \bar{v}) - \lambda I) = 0$$

THEOREM

Let λ_1 and λ_2 be the eigenvalues of $J(\bar{u}, \bar{v})$

- If $\operatorname{Re} \lambda_1 < 0$ and $\operatorname{Re} \lambda_2 < 0$ then the steady state $(\bar{u}, \bar{v})$ is **asymptotically stable**
- If $\operatorname{Re} \lambda_i > 0$ for at least one eigenvalue then the steady state $(\bar{u}, \bar{v})$ is **unstable**

If an eigenvalue has zero real part, linearization alone may be inconclusive

Eigenvalues

$$\begin{pmatrix} \partial_t \eta \\ \partial_t \xi \end{pmatrix} = J(\bar{u}, \bar{v}) \begin{pmatrix} \eta \\ \xi \end{pmatrix} \qquad J(\bar{u}, \bar{v}) = \begin{pmatrix} f_u(\bar{u}, \bar{v}) & f_v(\bar{u}, \bar{v}) \\ g_u(\bar{u}, \bar{v}) & g_v(\bar{u}, \bar{v}) \end{pmatrix}$$

CHARACTERISTIC EQUATION

$$\det(J - \lambda I) = \begin{vmatrix} f_u - \lambda & f_v \\ g_u & g_v - \lambda \end{vmatrix} = (f_u - \lambda)(g_v - \lambda) - f_v g_u = \lambda^2 - \mathrm{tr}(J)\lambda + \det(J)$$

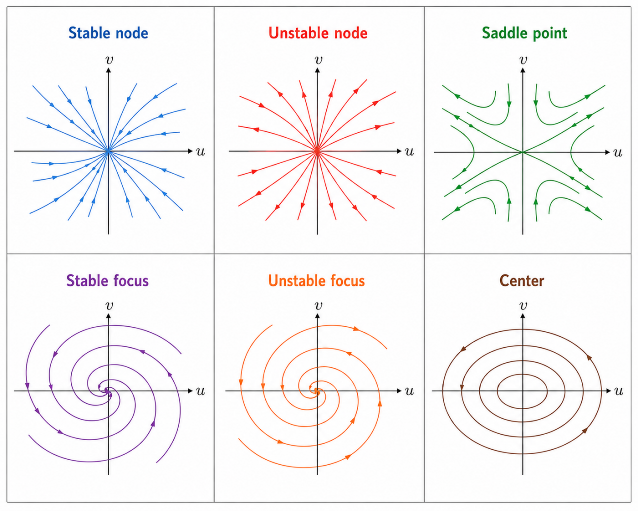
DETERMINANT

- If $\det(J) < 0$, one eigenvalue is positive and one is negative
  The steady state is an unstable saddle
- If $\det(J) > 0$, real eigenvalues have the same sign or form a complex conjugate pair
  The trace determines the sign of their real parts

TRACE WHEN $\det(J) > 0$

- If $\mathrm{tr}(J) < 0$, both eigenvalues have negative real parts
  The steady state is asymptotically stable
- If $\mathrm{tr}(J) > 0$, both eigenvalues have positive real parts
  The steady state is unstable

Eigenvalues determine local stability



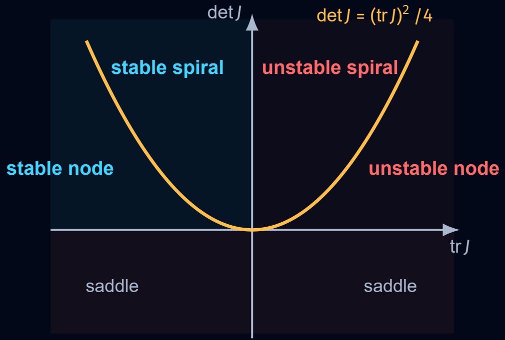
Trace and determinant classify a two-variable equilibrium

$$\det(J(\bar{u}, \bar{v}) - \lambda I) = 0$$

$$\boxed{\lambda^2 - \mathrm{tr}(J)\lambda + \det(J) = 0}$$

Therefore

$$\begin{aligned} \lambda_1 + \lambda_2 &= \mathrm{tr}(J) \\ \lambda_1 \lambda_2 &= \det(J) \end{aligned}$$



LECTURE MAP

The road from Hydra to pattern formation

From biological observations to mathematical mechanisms.

01 The biological puzzle

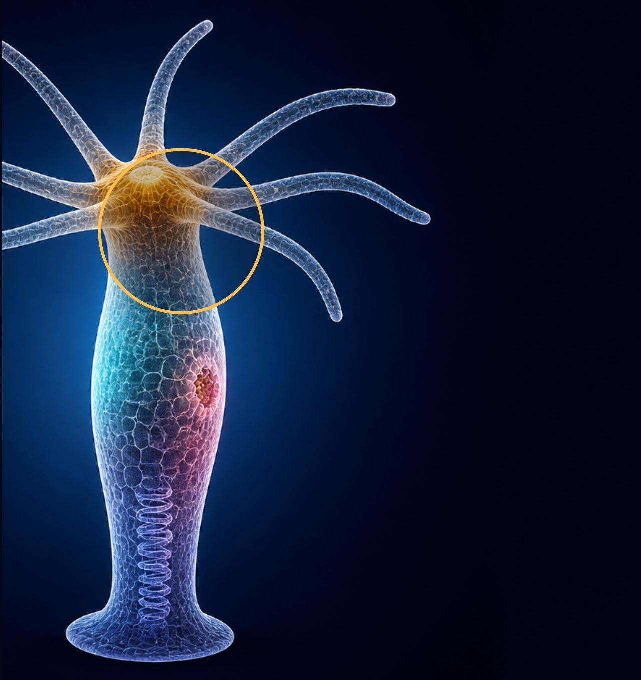
02 Reaction–diffusion approach

03 Turing instability

04 Gierer–Meinhardt model

05 Growth, injury & source density

06 Mechanochemical pattern formation



03 TURING INSTABILITY

The Turing idea

Two interacting substances can turn diffusion from a smoothing process into a source of spatial structure

Scalar reaction–diffusion equation: no stable patterns

$$
\begin{aligned}
\partial_t u &= D\Delta u + f(u) && x \in \Omega \quad t > 0 \\
u(x,0) &= u_0(x) && x \in \Omega \\
\nabla u \cdot \mathbf{n} &= 0 && x \in \partial\Omega
\end{aligned}
$$

THEOREM

Let $\Omega \subset \mathbb{R}^n$ be a bounded convex domain. Every nonconstant stationary solution of the scalar reaction–diffusion equation with homogeneous Neumann boundary conditions is unstable

Every stable stationary solution is spatially constant

$$u(x) \equiv \bar{u} \qquad \text{and} \qquad f(\bar{u}) = 0$$

Diffusion-driven instability

1952

Alan Turing

The Chemical Basis of Morphogenesis

$$\partial_t u = D_u \Delta u + f(u,v)$$
$$\partial_t v = D_v \Delta v + g(u,v)$$

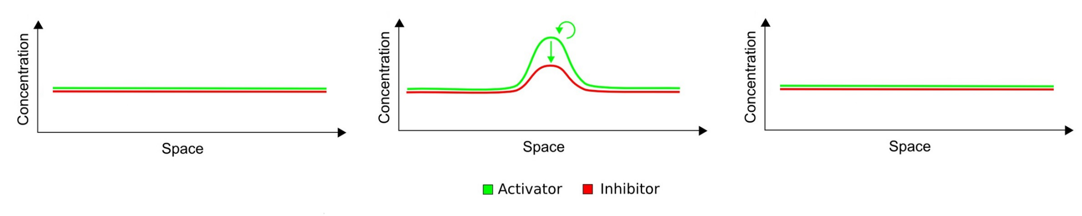
- **Stable reactions**
  Without spatial transport, small perturbations return to the homogeneous equilibrium.
- **Unstable reaction–diffusion system**
  After diffusion is introduced, one nonconstant spatial mode can begin to grow.

$$\boxed{D_u \neq D_v}$$

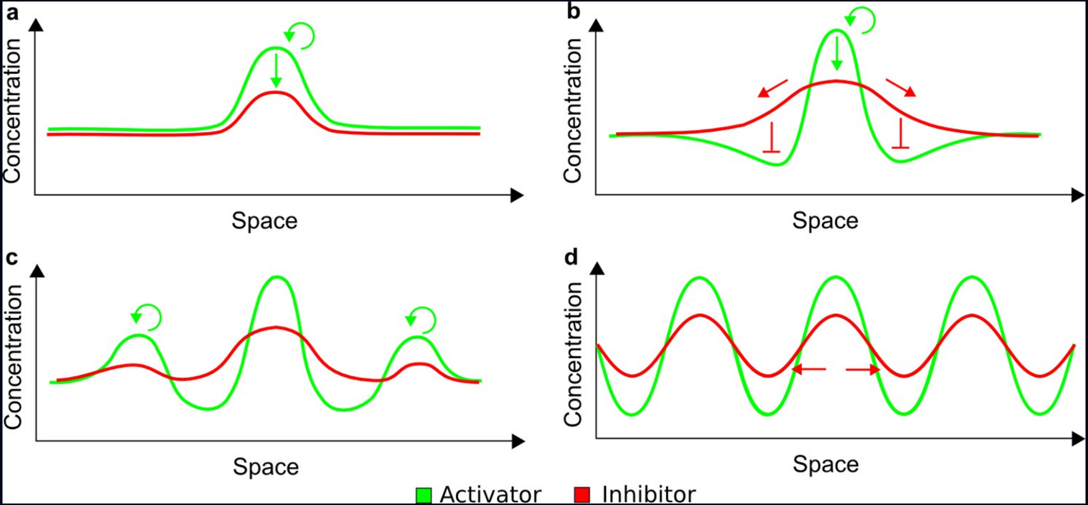
Unequal diffusion is essential for classical two-species Turing instability.

A. M. Turing, *Phil. Trans. R. Soc. B* 237 (1952), 37–72. Portrait: Elliott & Fry, 1951, Wikimedia Commons.

Equal diffusion shifts every spatial mode toward stability



Unequal diffusion can amplify a spatial mode



Stability analysis

Linearization and spatial eigenfunctions reduce the PDE to one stability problem for each spatial mode

The full reaction–diffusion problem

Let $\Omega \subset \mathbb{R}^n$ be a bounded domain

$$\partial_t u = D_u \Delta u + f(u,v) \qquad x \in \Omega \quad t > 0$$
$$\partial_t v = D_v \Delta v + g(u,v) \qquad x \in \Omega \quad t > 0$$

Initial conditions

$$u(x,0) = u_0(x) \qquad v(x,0) = v_0(x) \qquad x \in \Omega$$

Homogeneous Neumann boundary conditions

$$\nabla u \cdot \mathbf{n} = 0 \qquad \nabla v \cdot \mathbf{n} = 0 \qquad x \in \partial\Omega$$

- $f, g \in C^1(\mathbb{R}^2)$ describe local production, degradation and interactions at each spatial position
- $D_u, D_v > 0$ are the diffusion coefficients

Homogeneous steady state

HOMOGENEOUS EQUILIBRIUM

$$\begin{pmatrix} u(x,t) \\ v(x,t) \end{pmatrix} = \begin{pmatrix} \bar{u} \\ \bar{v} \end{pmatrix}$$

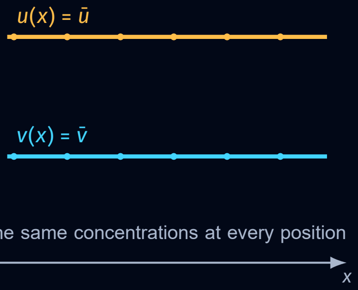
$$f(\bar{u},\bar{v}) = 0 \qquad g(\bar{u},\bar{v}) = 0$$

LOCAL JACOBIAN

$$J(\bar{u},\bar{v}) = \begin{pmatrix} f_u(\bar{u},\bar{v}) & f_v(\bar{u},\bar{v}) \\ g_u(\bar{u},\bar{v}) & g_v(\bar{u},\bar{v}) \end{pmatrix}$$

Both eigenvalues of $J(\bar{u},\bar{v})$ have negative real parts

$$\boxed{\operatorname{tr} J(\bar{u},\bar{v}) < 0 \qquad \det J(\bar{u},\bar{v}) > 0}$$

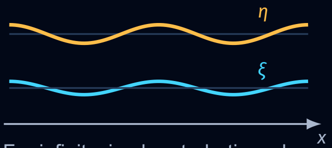


- **Stable without spatial variation**
  The reaction kinetics restore every sufficiently small homogeneous perturbation.

Perturb both concentrations around the steady state

SMALL PERTURBATION

$$u(x,t) = \bar{u} + \eta(x,t)$$
$$v(x,t) = \bar{v} + \xi(x,t)$$



For infinitesimal perturbations, keep only first-order terms

Expand both reaction terms around $(\bar{u},\bar{v})$

$$f(\bar{u}+\eta,\bar{v}+\xi) = f(\bar{u},\bar{v}) + f_u(\bar{u},\bar{v})\eta + f_v(\bar{u},\bar{v})\xi + O\big(\|(\eta,\xi)\|^2\big)$$
$$g(\bar{u}+\eta,\bar{v}+\xi) = g(\bar{u},\bar{v}) + g_u(\bar{u},\bar{v})\eta + g_v(\bar{u},\bar{v})\xi + O\big(\|(\eta,\xi)\|^2\big)$$

Since $(\bar{u},\bar{v})$ is constant and $f(\bar{u},\bar{v}) = g(\bar{u},\bar{v}) = 0$, substitution gives

$$\partial_t\eta = D_u\Delta\eta + f_u(\bar{u},\bar{v})\eta + f_v(\bar{u},\bar{v})\xi + O\big(\|(\eta,\xi)\|^2\big)$$
$$\partial_t\xi = D_v\Delta\xi + g_u(\bar{u},\bar{v})\eta + g_v(\bar{u},\bar{v})\xi + O\big(\|(\eta,\xi)\|^2\big)$$

$$\boxed{\partial_t\begin{pmatrix}\eta\\ \xi\end{pmatrix} = J(\bar{u},\bar{v})\begin{pmatrix}\eta\\ \xi\end{pmatrix} + \begin{pmatrix}D_u & 0\\ 0 & D_v\end{pmatrix}\Delta\begin{pmatrix}\eta\\ \xi\end{pmatrix}}$$

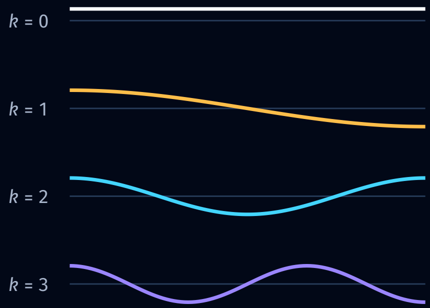
Spatial eigenfunctions separate the perturbation

Use the eigenfunctions introduced in the diffusion section

$$-\Delta\phi_k = \mu_k\phi_k \qquad \nabla\phi_k \cdot n = 0 \quad \text{on } \partial\Omega$$

$$\begin{pmatrix}\eta(x,t)\\ \xi(x,t)\end{pmatrix} = \sum_{k=0}^{\infty}\begin{pmatrix}a_k(t)\\ b_k(t)\end{pmatrix}\phi_k(x)$$

- **One spatial shape**
  $\phi_k(x)$ determines where the perturbation is positive and negative.
- **Two chemical amplitudes**
  $a_k(t)$ and $b_k(t)$ determine how strongly that shape appears in each substance.



Each spatial shape will evolve independently after linearization.

Each mode is a two-dimensional stability problem

Combining the linearized perturbation equation with the eigenfunction expansion

$$\partial_t \begin{pmatrix} \eta \\ \xi \end{pmatrix} = J(\bar{u},\bar{v}) \begin{pmatrix} \eta \\ \xi \end{pmatrix} + \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix} \Delta \begin{pmatrix} \eta \\ \xi \end{pmatrix} \quad + \quad \begin{pmatrix} \eta(x,t) \\ \xi(x,t) \end{pmatrix} = \sum_{k=0}^{\infty} \begin{pmatrix} a_k(t) \\ b_k(t) \end{pmatrix} \phi_k(x)$$

$$\Downarrow$$

$$\sum_{k=0}^{\infty} \frac{d}{dt} \begin{pmatrix} a_k \\ b_k \end{pmatrix} \phi_k = J(\bar{u},\bar{v}) \sum_{k=0}^{\infty} \begin{pmatrix} a_k \\ b_k \end{pmatrix} \phi_k + \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix} \sum_{k=0}^{\infty} \begin{pmatrix} a_k \\ b_k \end{pmatrix} \Delta \phi_k$$

Since $\Delta\phi_k = -\mu_k \phi_k$ and $\{\phi_k\}_{k=0}^{\infty}$ is an orthonormal basis, the coefficients of every ϕ_k must agree

$$\boxed{\frac{d}{dt} \begin{pmatrix} a_k \\ b_k \end{pmatrix} = \left(J(\bar{u},\bar{v}) - \mu_k \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix} \right) \begin{pmatrix} a_k \\ b_k \end{pmatrix}}$$

Define the stability matrix of each mode

$$\frac{d}{dt}\begin{pmatrix} a_k \\ b_k \end{pmatrix} = \left(J(\bar{u}, \bar{v}) - \mu_k \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix}\right)\begin{pmatrix} a_k \\ b_k \end{pmatrix}$$

Define the stability matrix associated with the kth spatial mode

$$\boxed{M_k = J(\bar{u}, \bar{v}) - \mu_k \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix} = \begin{pmatrix} f_u(\bar{u}, \bar{v}) - D_u \mu_k & f_v(\bar{u}, \bar{v}) \\ g_u(\bar{u}, \bar{v}) & g_v(\bar{u}, \bar{v}) - D_v \mu_k \end{pmatrix}}$$

$$\boxed{\frac{d}{dt}\begin{pmatrix} a_k \\ b_k \end{pmatrix} = M_k \begin{pmatrix} a_k \\ b_k \end{pmatrix}}$$

Every spatial wavelength has its own 2×2 stability matrix

The homogeneous mode remains locally stable

HOMOGENEOUS MODE

$$\phi_0(x) = \text{constant} \qquad \mu_0 = 0$$

$$\boxed{M_0 = J(\bar{u}, \bar{v})}$$

the same perturbation everywhere

- **Stable by assumption**
  The local reaction analysis already guarantees decay of this mode.

NONCONSTANT MODES

$$\mu_k > 0 \qquad k \geq 1$$

$$M_k = J(\bar{u}, \bar{v}) - \mu_k \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix}$$

a spatial pattern can change stability

- **The Turing question**
  Can any M_k acquire an eigenvalue with positive real part?

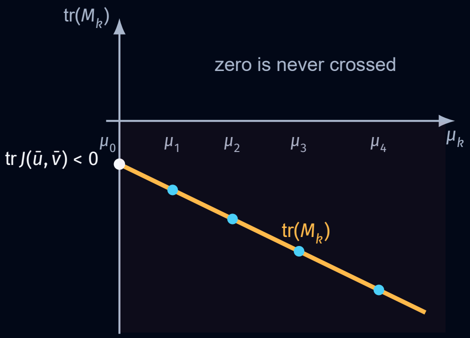
Diffusion makes the trace more negative

$$M_k = \begin{pmatrix} f_u(\bar{u},\bar{v}) - D_u\mu_k & f_v(\bar{u},\bar{v}) \\ g_u(\bar{u},\bar{v}) & g_v(\bar{u},\bar{v}) - D_v\mu_k \end{pmatrix}$$

$$\boxed{\mathrm{tr}(M_k) = \mathrm{tr}\,J(\bar{u},\bar{v}) - (D_u + D_v)\mu_k}$$

- **The trace starts negative**
  $\mu_0 = 0$ and $\mathrm{tr}(M_0) = \mathrm{tr}\,J(\bar{u},\bar{v}) < 0$ because the homogeneous equilibrium is locally stable.
- **Diffusion decreases it further**
  For every $k \geq 1$, the term $-(D_u + D_v)\mu_k$ is strictly negative.

Therefore a Turing instability cannot occur through the trace becoming positive.



The determinant contains the instability mechanism

Compute the determinant of the mode matrix

$$\begin{aligned}\det(M_k) &= \det\begin{pmatrix} f_u(\bar{u},\bar{v}) - D_u\mu_k & f_v(\bar{u},\bar{v}) \\ g_u(\bar{u},\bar{v}) & g_v(\bar{u},\bar{v}) - D_v\mu_k \end{pmatrix} \\ &= \left(f_u(\bar{u},\bar{v}) - D_u\mu_k\right)\left(g_v(\bar{u},\bar{v}) - D_v\mu_k\right) - f_v(\bar{u},\bar{v})g_u(\bar{u},\bar{v}) \\ &= \det J(\bar{u},\bar{v}) - \left(D_v f_u(\bar{u},\bar{v}) + D_u g_v(\bar{u},\bar{v})\right)\mu_k + D_u D_v \mu_k^2\end{aligned}$$

$$\boxed{\det(M_k) = \det J(\bar{u},\bar{v}) - A\mu_k + D_u D_v \mu_k^2} \qquad A = D_v f_u(\bar{u},\bar{v}) + D_u g_v(\bar{u},\bar{v})$$

- **Positive at $\mu_0 = 0$**
  $\det(M_0) = \det J(\bar{u},\bar{v}) > 0$ follows from local stability.
- **The linear term can decrease it**
  Unequal diffusion weights f_u and g_v differently inside A.
- **Very short waves are stable**
  $D_u D_v \mu_k^2$ dominates for sufficiently large k.

Turing instability condition

$$\boxed{\det(M_k) = \det J(\bar{u},\bar{v}) - A\mu_k + D_u D_v \mu_k^2}$$

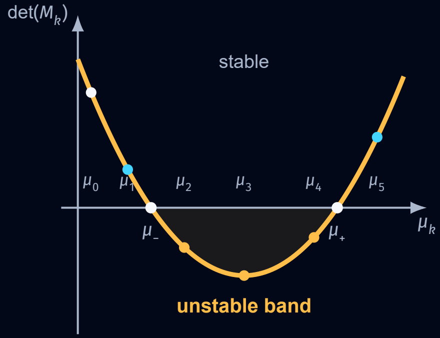
- $\det(M_k) > 0$
  The kth spatial mode is stable.
- $\det(M_k) < 0$
  The kth spatial mode is unstable.

$$\det(M_k) < 0 \iff \mu_- < \mu_k < \mu_+$$

A negative interval exists when

$$A > 0 \qquad A^2 > 4D_u D_v \det J(\bar{u},\bar{v})$$

$$\mu_\pm = \frac{A \pm \sqrt{A^2 - 4D_u D_v \det J(\bar{u},\bar{v})}}{2D_u D_v}$$



Turing instability theorem

$$\boxed{\begin{aligned} \partial_t u &= D_u \Delta u + f(u,v) \\ \partial_t v &= D_v \Delta v + g(u,v) \end{aligned}}$$

THEOREM

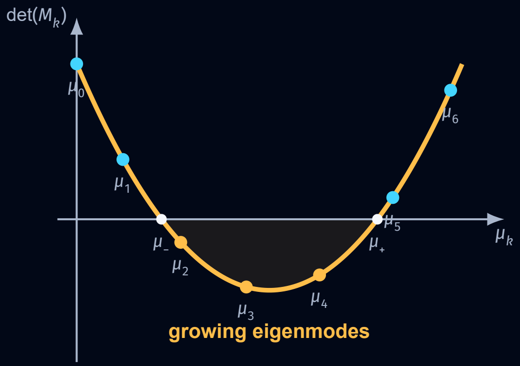
If the system is stable in the absence of diffusion, and u satisfies the autocatalysis condition, i.e. $f_u(\bar{u},\bar{v}) > 0$, diffusion D_u is sufficiently small and diffusion D_v is sufficiently large, then diffusion destabilizes the homogeneous steady state $(\bar{u},\bar{v})$.

03 TURING INSTABILITY

Mode selection

The unstable band must intersect the discrete spatial spectrum allowed by the tissue geometry

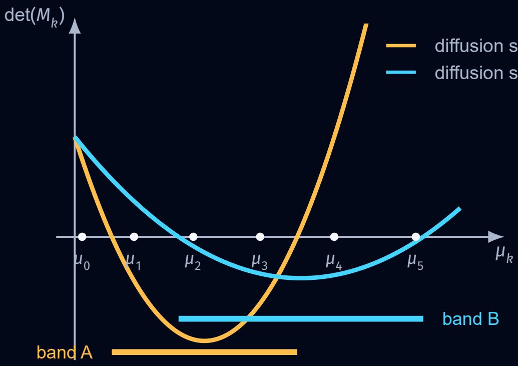
Only eigenvalues inside the unstable band can grow



- **Only selected modes grow**
  The unstable band is continuous, but the tissue supplies only discrete eigenvalues.

$\det(M_k) < 0$ $\Longrightarrow$ Exists a positive eigenvalue $\lambda_{k,+}$ $\Longrightarrow$ Spatial mode k grows at rate $e^{\lambda_{k,+}t}$

Diffusion changes which modes are unstable



$$\det(M_k) = \det J(\bar{u}, \bar{v}) - A\mu_k + D_u D_v \mu_k^2$$

- **The band moves**
  Changing D_u and D_v changes its location and width.
- **The selected modes change**
  A different set of the fixed eigenvalues μ_k can become unstable.

Diffusion therefore controls the preferred spatial scale.

Diffusion and domain size

Let $\tilde{\Omega}$ be a fixed reference domain and scale it by $L > 0$

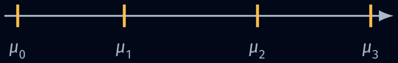
$$\Omega_L = L\tilde{\Omega} \qquad x = L\tilde{x}$$

Under the change of variables $x = L\tilde{x}$, the system is equivalently written on the reference domain as

$$\partial_t \tilde{u} = \frac{D_u}{L^2}\Delta_{\tilde{x}}\tilde{u} + f(\tilde{u},\tilde{v})$$
$$\partial_t \tilde{v} = \frac{D_v}{L^2}\Delta_{\tilde{x}}\tilde{v} + g(\tilde{u},\tilde{v})$$

$$L\tilde{\Omega} \textbf{ with } (D_u, D_v) \quad \Leftrightarrow \quad \tilde{\Omega} \textbf{ with } \left(\frac{D_u}{L^2}, \frac{D_v}{L^2}\right)$$

small domain



large domain

increasing L compresses the spatial spectrum

- **Important distinction**
  Changing the ratio D_v / D_u is not equivalent to changing the domain size.

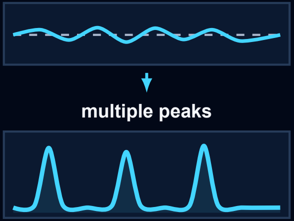
A larger domain can support more peaks

SMALL DOMAIN

homogeneous steady state with a small perturbation

LARGER DOMAIN

homogeneous steady state with a small perturbation



- **Linear level**
  Turing analysis identifies which modes lie in the unstable band and can grow
- **Pattern formation requires nonlinearity**
  Nonlinear interactions select and saturate the emerging spatial pattern
- **Useful but incomplete prediction**
  Unstable eigenmodes can be useful for predicting the pattern, but they do not provide a complete explanation

LECTURE MAP

The road from Hydra to pattern formation

From biological observations to mathematical mechanisms.

Return to the Gierer–Meinhardt model

Let $\Omega \subset \mathbb{R}^n$ be a bounded domain

$$\partial_t u = D_u \Delta u + \frac{au^2}{K + v} - \mu_u u$$
$$\partial_t v = D_v \Delta v + bu^2 - \mu_v v$$

Initial conditions

$$u(x, 0) = u_0(x) \qquad v(x, 0) = v_0(x) \qquad x \in \Omega$$

Homogeneous Neumann boundary conditions

$$\nabla u \cdot \mathbf{n} = 0 \qquad \nabla v \cdot \mathbf{n} = 0 \qquad x \in \partial\Omega$$

- **Homogeneous states**
  Which constant concentrations can persist in time?
- **Local stability**
  Which states recover after a spatially uniform perturbation?
- **Diffusion-driven instability**
  Can diffusion destabilize a locally stable active state?

u activator concentration

v inhibitor concentration

D_u, D_v diffusion coefficients

a activator self-amplification

K inhibition scale

b inhibitor production

μ_u, μ_v degradation rates

04 GIERER–MEINHARDT MODEL

Steady states

The geometry of the local reactions reveals an activation threshold and two competing stable outcomes

Reaction system

Looking for steady states and analyzing stability

$$\partial_t u = f(u,v) = \frac{au^2}{K+v} - \mu_u u$$
$$\partial_t v = g(u,v) = bu^2 - \mu_v v$$

Without diffusion every spatial position follows the same two-variable reaction system

- **Steady states**
  Find constant concentrations $(\bar{u}, \bar{v})$ for which $f(\bar{u}, \bar{v}) = 0$ and $g(\bar{u}, \bar{v}) = 0$
- **Local stability**
  Linearize the reaction system around each steady state and inspect the eigenvalues of the Jacobian

Steady states

A homogeneous steady state $(\bar{u}, \bar{v})$ satisfies

$$0 = \frac{a\bar{u}^2}{K + \bar{v}} - \mu_u \bar{u}$$
$$0 = b\bar{u}^2 - \mu_v \bar{v}$$

The second equation gives

$$\bar{v} = \frac{b}{\mu_v}\bar{u}^2$$

Substitution into the first equation gives

$$0 = \frac{a\bar{u}^2}{K + \frac{b}{\mu_v}\bar{u}^2} - \mu_u \bar{u}$$

After clearing the denominator

$$\boxed{\bar{u}\left(\frac{b}{\mu_v}\bar{u}^2 - \frac{a}{\mu_u}\bar{u} + K\right) = 0}$$

A cubic equation determines all homogeneous steady states

Steady states: two positive roots

$$\bar{u} = 0 \quad \Rightarrow \quad (\bar{u}, \bar{v}) = (0, 0)$$

$$\bar{u} > 0 \quad \Rightarrow \quad \frac{b}{\mu_v}\bar{u}^2 - \frac{a}{\mu_u}\bar{u} + K = 0$$

TWO POSITIVE STEADY STATES IF $a^2\mu_v > 4bK\mu_u^2$

$$u_\pm = \frac{\mu_v}{2b}\left[\frac{a}{\mu_u} \pm \sqrt{\left(\frac{a}{\mu_u}\right)^2 - \frac{4bK}{\mu_v}}\right]$$

$$v_\pm = \frac{b}{\mu_v}u_\pm^2$$

- **Lower state** (u_-, v_-)
  The smaller positive root will become the activation threshold
- **Upper state** (u_+, v_+)
  The larger positive root is the candidate active state

The nonlinear reaction terms can create two distinct positive concentration levels

The nullclines

ACTIVATOR NULLCLINE

$$f(u,v) = \frac{au^2}{K+v} - \mu_u u = 0$$

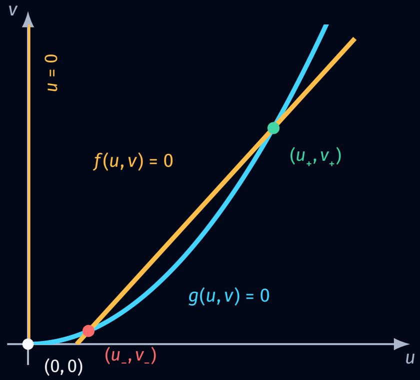
$$\iff \quad u = 0 \quad \text{or} \quad v = \frac{a}{\mu_u}u - K$$

INHIBITOR NULLCLINE

$$g(u,v) = bu^2 - \mu_v v = 0$$

$$\iff \quad v = \frac{b}{\mu_v}u^2$$

Positive equilibria are intersections of a straight line and a parabola



Possible steady states

EXISTENCE CONDITION

$$\boxed{\left(\frac{a}{\mu_u}\right)^2 > \frac{4bK}{\mu_v}}$$

At equality, the line is tangent to the parabola and the two positive equilibria merge

Strong enough autocatalysis creates the zero state and two positive states separated by a threshold

04 GIERER–MEINHARDT MODEL

Local stability

We first analyse stability in the local reaction system without diffusion and identify the activation threshold

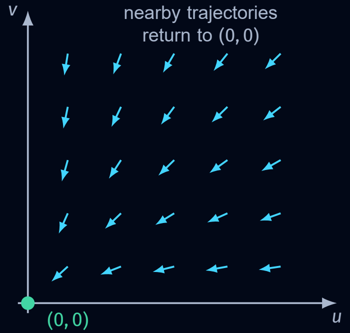
The zero steady state is always stable

$$f(u,v) = \frac{au^2}{K+v} - \mu_u u \qquad g(u,v) = bu^2 - \mu_v v$$

JACOBIAN

$$J(u,v) = \begin{pmatrix} f_u(u,v) & f_v(u,v) \\ g_u(u,v) & g_v(u,v) \end{pmatrix} = \begin{pmatrix} \frac{2au}{K+v} - \mu_u & -\frac{au^2}{(K+v)^2} \\ 2bu & -\mu_v \end{pmatrix}$$

$$J(0,0) = \begin{pmatrix} -\mu_u & 0 \\ 0 & -\mu_v \end{pmatrix}, \qquad \lambda_1 = -\mu_u < 0, \quad \lambda_2 = -\mu_v < 0$$



- **Asymptotically stable**
  Near $u = 0$ quadratic activator production is dominated by linear degradation

Jacobian at the positive steady states

At either positive steady state $(u_\pm, v_\pm)$ the activator equation gives

$$f(u_\pm, v_\pm) = \frac{au_\pm^2}{K + v_\pm} - \mu_u u_\pm = 0 \quad \Rightarrow \quad \frac{au_\pm}{K + v_\pm} = \mu_u$$

ACTIVATOR DERIVATIVES

$$f_u(u_\pm, v_\pm) = \frac{2au_\pm}{K + v_\pm} - \mu_u = \mu_u$$

$$f_v(u_\pm, v_\pm) = -\frac{au_\pm^2}{(K + v_\pm)^2} = -\frac{\mu_u^2}{a}$$

INHIBITOR DERIVATIVES

$$g_u(u_\pm, v_\pm) = 2bu_\pm$$

$$g_v(u_\pm, v_\pm) = -\mu_v$$

$$\boxed{J(u_\pm, v_\pm) = \begin{pmatrix} \mu_u & -\mu_u^2/a \\ 2bu_\pm & -\mu_v \end{pmatrix}}$$

The two positive steady states differ only through the entry $2bu_\pm$

The lower positive steady state is a saddle

$$\operatorname{tr} J(u_\pm, v_\pm) = \mu_u - \mu_v$$

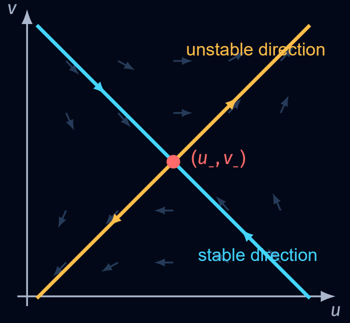
$$\det J(u_\pm, v_\pm) = \mu_u \left(\frac{2b\mu_u}{a} u_\pm - \mu_v \right)$$

The two positive roots lie on opposite sides of their midpoint

$$u_- < \frac{a\mu_v}{2b\mu_u} < u_+$$

$$\boxed{\det J(u_-, v_-) < 0} \quad \Rightarrow \quad \lambda_1 \lambda_2 < 0$$

- **Saddle**
  One eigenvalue is negative and the other is positive



The upper positive steady state can be stable

$$u_+ > \frac{a\mu_v}{2b\mu_u}$$

$$\det J(u_+, v_+) > 0$$

$$\operatorname{tr} J(u_+, v_+) = \mu_u - \mu_v$$

$$\boxed{\mu_u < \mu_v} \quad \Rightarrow \quad \operatorname{tr} J(u_+, v_+) < 0$$

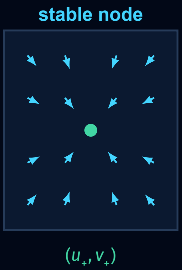
$$\Rightarrow \quad \operatorname{Re}\lambda_1 < 0 \qquad \operatorname{Re}\lambda_2 < 0$$

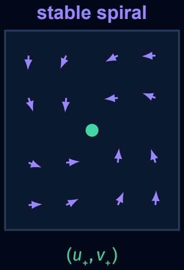
The discriminant determines the local geometry

$$D_+ = \left(\operatorname{tr} J(u_+, v_+)\right)^2 - 4 \det J(u_+, v_+)$$

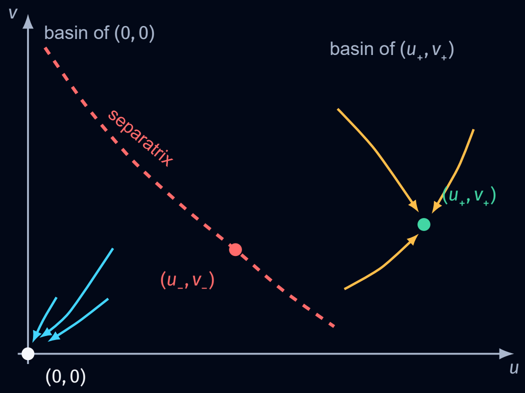
$$D_+ \geq 0 \quad \Rightarrow \quad \text{stable node}$$

$$D_+ < 0 \quad \Rightarrow \quad \text{stable spiral}$$





The saddle separates the two stable outcomes



- **Small perturbations**
  States below the threshold return to $(0, 0)$
- **Activation threshold**
  The stable manifold of (u_-, v_-) forms the basin boundary
- **Large perturbations**
  States beyond the threshold approach (u_+, v_+)

The final state depends on whether the perturbation crosses the nonlinear threshold

04 GIERER–MEINHARDT MODEL

Turing instability

Diffusion acts on the stable positive state and selects a band of growing spatial eigenmodes

Choose the homogeneous steady state

CONSTANT POSITIVE STATE

$$\begin{pmatrix} u(x,t) \\ v(x,t) \end{pmatrix} = \begin{pmatrix} u_+ \\ v_+ \end{pmatrix} \qquad x \in \Omega$$

The same concentrations occur at every spatial position

STABLE WITHOUT DIFFUSION

$$\operatorname{tr} J(u_+, v_+) = \mu_u - \mu_v < 0$$
$$\det J(u_+, v_+) > 0$$

$$\boxed{\mu_u < \mu_v}$$

$$J(u_+, v_+) = \begin{pmatrix} \mu_u & -\mu_u^2/a \\ 2bu_+ & -\mu_v \end{pmatrix}$$

Turing theorem applies to Gierer-Meinhardt

$$J(u_+, v_+) = \begin{pmatrix} \mu_u & -\mu_u^2/a \\ 2bu_+ & -\mu_v \end{pmatrix} \qquad f_u(u_+, v_+) = \mu_u > 0$$

THEOREM

If a two-variable reaction system is stable without diffusion and $f_u(\bar{u}, \bar{v}) > 0$ then D_u can be chosen sufficiently small and D_v sufficiently large so that diffusion destabilizes the homogeneous steady state

Suitable diffusion coefficients can destabilize the stable positive GM steady state

Each spatial mode gives a two-dimensional system

SPATIAL EIGENFUNCTION

$$-\Delta\phi_k = \mu_k \phi_k$$

$$\mu_k \longrightarrow \infty \qquad k \longrightarrow \infty$$

Perturb both concentrations in the same spatial mode

$$u(x,t) = u_+ + \alpha_k(t)\phi_k(x)$$
$$v(x,t) = v_+ + \beta_k(t)\phi_k(x)$$

AMPLITUDE EQUATION

$$\frac{d}{dt}\begin{pmatrix}\alpha_k \\ \beta_k\end{pmatrix} = M_k \begin{pmatrix}\alpha_k \\ \beta_k\end{pmatrix}$$

$$\boxed{M_k = J(u_+, v_+) - \mu_k \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix} = \begin{pmatrix} \mu_u - D_u\mu_k & -\mu_u^2/a \\ 2bu_+ & -\mu_v - D_v\mu_k \end{pmatrix}}$$

Every eigenmode has its own explicit 2×2 **stability matrix**

Exact diffusion conditions for Turing instability

$$M_k = \begin{pmatrix} \mu_u - D_u \mu_k & -\mu_u^2/a \\ 2bu_+ & -\mu_v - D_v \mu_k \end{pmatrix}$$

$$\boxed{\det(M_k) = \det J(u_+, v_+) - A_{\mathrm{GM}} \mu_k + D_u D_v \mu_k^2} \qquad A_{\mathrm{GM}} = D_v \mu_u - D_u \mu_v$$

- **The parabola must initially decrease**
  $A_{\mathrm{GM}} > 0$ is equivalent to faster inhibitor diffusion

$$A_{\mathrm{GM}} > 0 \quad \Longleftrightarrow \quad \boxed{\frac{D_v}{D_u} > \frac{\mu_v}{\mu_u}}$$

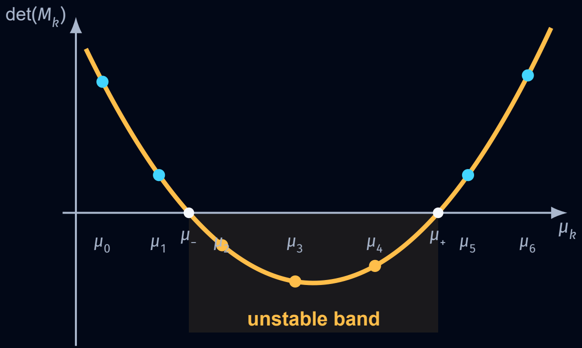
- **The minimum must cross zero**
  The quadratic must have two distinct positive roots

$$A_{\mathrm{GM}}^2 > 4 D_u D_v \det J(u_+, v_+)$$

$$\mu_\pm = \frac{A_{\mathrm{GM}} \pm \sqrt{A_{\mathrm{GM}}^2 - 4 D_u D_v \det J(u_+, v_+)}}{2 D_u D_v}$$

$$\det(M_k) < 0 \quad \Longleftrightarrow \quad \mu_- < \mu_k < \mu_+$$

Only GM eigenmodes inside the unstable band can grow



- **Unstable modes**
  Only discrete eigenvalues μ_k between μ_- and μ_+ can grow
- **Stable modes**
  Modes outside the negative determinant interval decay

$$\det(M_k) < 0 \quad \Rightarrow \quad \lambda_{k,+} > 0 \quad \Rightarrow \quad (\alpha_k(t), \beta_k(t)) \sim e^{\lambda_{k,+} t}$$

04 GIERER–MEINHARDT MODEL

Example

Certain set of parameters

Example

Let $\Omega = [0, 1] \subset \mathbb{R}$ be an interval

$$\partial_t u = 0.01\Delta u + \frac{1.5u^2}{1 + v} - 0.5u$$
$$\partial_t v = \Delta v + 2u^2 - v$$

PARAMETER VALUES

$D_u = 0.01$	$D_v = 1$
$a = 1.5$	$b = 2$
$\mu_u = 0.5$	$\mu_v = 1$
$K = 1$	

- **Initial conditions**
  $u(x, 0) = u_0(x)$ and $v(x, 0) = v_0(x)$
- **No-flux boundaries**
  $\partial_x u(x, t) = \partial_x v(x, t) = 0$ at $x = 0$ and $x = 1$

Three steady states

Constant steady states solve

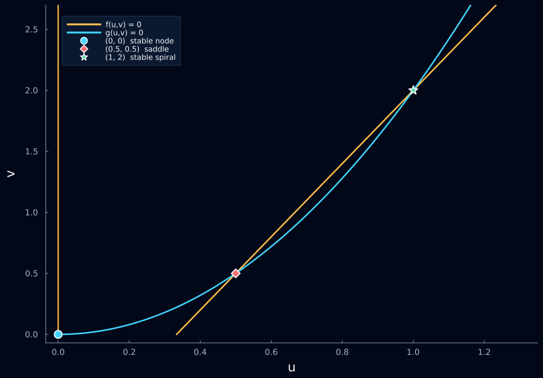
$$0 = \frac{1.5\bar{u}^2}{1 + \bar{v}} - 0.5\bar{u} \qquad\qquad 0 = 2\bar{u}^2 - \bar{v}$$

$$\bar{v} = 2\bar{u}^2$$

$$\bar{u}\left(2\bar{u}^2 - 3\bar{u} + 1\right) = \bar{u}\,(2\bar{u} - 1)(\bar{u} - 1) = 0$$

$$(\bar{u}, \bar{v}) = (0, 0) \qquad (\bar{u}, \bar{v}) = (0.5, 0.5) \qquad (\bar{u}, \bar{v}) = (1, 2)$$

Nullclines



- **Activator nullcline**
  $u = 0$ or $v = 3u - 1$
- **Inhibitor nullcline**
  $v = 2u^2$
- **Three intersections**
  Every intersection is a steady state of the local reaction system

Steady states: local classification

Evaluate the Jacobian and its eigenvalues at each steady state

$(\bar{u}, \bar{v}) = (0, 0)$

$\lambda_1 = -1$

$\lambda_2 = -0.5$

- **Stable node**
  Both eigenvalues are negative

$(\bar{u}, \bar{v}) = (0.5, 0.5)$

$\lambda_- = -0.72871$

$\lambda_+ = 0.22871$

- **Saddle**
  One eigenvalue is positive

$(\bar{u}, \bar{v}) = (1, 2)$

$\lambda_\pm = -0.25 \pm 0.32275i$

- **Stable spiral**
  Both real parts are negative

Only the middle steady state has a positive local eigenvalue

Diffusion destabilizes the upper steady state

Linearization at the stable steady state $(1, 2)$

$$J(1,2) = \begin{pmatrix} 0.5 & -1/6 \\ 4 & -1 \end{pmatrix}$$

$$\operatorname{tr} J(1,2) = -0.5$$

$$\det J(1,2) = \frac{1}{6}$$

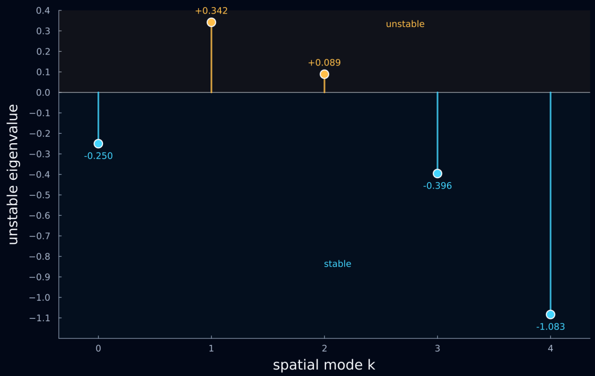
$$M_k = \begin{pmatrix} 0.5 - 0.01\mu_k & -1/6 \\ 4 & -1 - \mu_k \end{pmatrix}$$

$$\det(M_k) = \frac{1}{6} - 0.49\mu_k + 0.01\mu_k^2$$

$$\boxed{\det(M_k) < 0 \quad \Longleftrightarrow \quad 0.34253 < \mu_k < 48.65747}$$

Modes whose Laplacian eigenvalues lie inside this interval are unstable

Unstable eigenmodes



$D_u = 0.01$ $D_v = 1$ $L = 1$

- **Dominant mode $k = 1$**
  $\lambda_{1,+} = 0.34184$
- **Slower mode $k = 2$**
  $\lambda_{2,+} = 0.08878$
- **Stable neighbors**
  $k = 0, 3, \ldots$ decay

The first five spatial eigenmodes on $\Omega = [0, 1]$

Steady states of the local reaction system

$$(0, 0) \qquad (0.5, 0.5) \qquad (1, 2)$$

$$\phi_k(x) = \cos(k\pi x)$$

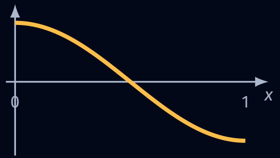
k	$\phi_k(x)$	ρ_k	behavior
1	$\cos(\pi x)$	+0.34184	unstable
2	$\cos(2\pi x)$	+0.08878	unstable
3	$\cos(3\pi x)$	−0.39572	stable
4	$\cos(4\pi x)$	−1.08336	stable
5	$\cos(5\pi x)$	−1.97011	stable

LINEARIZATION USED BELOW

$$(\bar{u}, \bar{v}) = (1, 2)$$

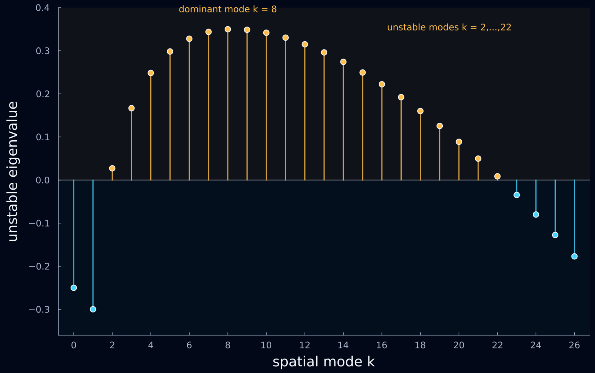
$$\rho_k = \max_{\lambda \in \sigma(M_k)} \operatorname{Re} \lambda$$

DOMINANT UNSTABLE MODE



$\phi_1(x) = \cos(\pi x)$

Unstable eigenmodes on a larger domain $\Omega = [0, 10]$



$D_u = 0.01, \quad D_v = 1, \quad L = 10$

- **Many unstable modes**
  $k = 2, \dots, 22$
- **Dominant mode $k = 8$**
  $\lambda_{8,+} = 0.34988$
- **Stable neighbors**
  $k = 0, 1, 23, \dots$ decay

The longer domain places more discrete eigenvalues inside the same unstable band

Selected spatial eigenmodes on $\Omega = [0, 10]$

$(0, 0)$ $(0.5, 0.5)$ $(1, 2)$

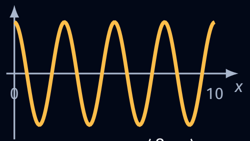
$(\bar{u}, \bar{v}) = (1, 2)$

$$\phi_k(x) = \cos\left(\frac{k\pi x}{10}\right)$$

$$\rho_k = \max_{\lambda \in \sigma(M_k)} \mathrm{Re}\,\lambda$$

k	$\phi_k(x)$	ρ_k	behavior
1	$\cos(\pi x/10)$	−0.29984	stable
2	$\cos(2\pi x/10)$	+0.02724	unstable
6	$\cos(6\pi x/10)$	+0.32788	unstable
7	$\cos(7\pi x/10)$	+0.34376	unstable
8	$\cos(8\pi x/10)$	+**0.34988**	unstable
9	$\cos(9\pi x/10)$	+0.34870	unstable
10	$\cos(10\pi x/10)$	+0.34184	unstable
21	$\cos(21\pi x/10)$	+0.04979	unstable
22	$\cos(22\pi x/10)$	+0.00864	unstable
23	$\cos(23\pi x/10)$	−0.03464	stable

DOMINANT UNSTABLE MODE

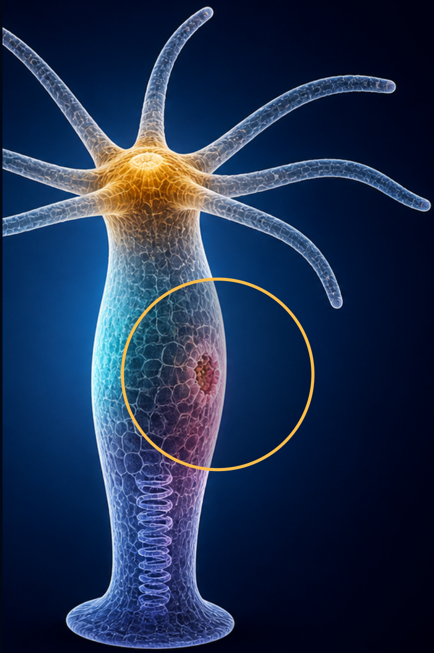


$$\phi_8(x) = \cos\left(\frac{8\pi x}{10}\right)$$

LECTURE MAP

The road from Hydra to pattern formation

From biological observations to mathematical mechanisms.



Can a local activator perturbation create a second organizer?

THE LIVE EXPERIMENT

- **Begin with one organizer**
  A stable peak is already established on the short domain.
- **Add activator locally**
  A brief, localized perturbation creates a second candidate peak away from the first.
- **Then release the system**
  After the perturbation, the autonomous nonlinear dynamics select the outcome.

WHAT COULD HAPPEN?

- **A second organizer forms**
  The added peak becomes self-maintaining and both organizers persist.
- **The added peak disappears**
  It remains transient, loses the competition and the original one-peak pattern returns.

Does local activation create a new organizer — or only a transient peak?

The small domain cannot sustain a second peak

NONLINEAR INTERPRETATION

$$(\bar{u}_1, \bar{v}_1) + (\delta u, 0) \quad \longrightarrow \quad (\bar{u}_1, \bar{v}_1).$$

- **Competition through inhibition**
  The broad inhibitor field couples the two candidate peaks.
- **Limited spatial capacity**
  For these parameters, [0, 1] supports the one-peak attractor but not two persistent organizers.

OBSERVED IN THE LIVE SIMULATION

- **The original peak persists**
  The established organizer remains self-maintaining.
- **The added peak decays**
  The local activator perturbation does not become a second organizer.

A local activator perturbation need not enter the basin of a new spatial pattern

Does a larger Hydra necessarily grow more heads?

MORE SPACE CHANGES WHAT CAN FIT

- **Peaks can separate**
  A longer domain creates regions farther from the existing organizer.
- **More structures are possible**
  Several well-separated activator peaks may now coexist.
- **Large fixed-domain experiment**
  Starting from a nearly uniform state with noise can produce many peaks.

BUT OUR HYDRA ALREADY HAS ONE HEAD

- **The initial state is different**
  Growth begins from an established organizer, not from uniform noise.
- **The real question**
  As the domain grows, will one head persist, split or generate additional heads?

A larger domain permits more peaks, growth determines whether they appear

Growth selects number of peaks

Let Ω grow in time: $\Omega(t) = [0, L(t)]$.

SLOW GROWTH

The domain length changes gradually.

the solution has time to reorganize as successive modes become available

FAST GROWTH

The domain length changes rapidly.

the domain crosses the same instability ranges on a different timescale

Same initial organizer. Same final length. Same final pattern?

The growth rate becomes a nonlinear pattern-selection parameter

05 GROWTH, INJURY & SOURCE DENSITY

Role of the wound

A cut changes the geometry,
but regeneration also requires
an injury-induced signal

Is removing the head enough to trigger regeneration?

WHAT THE LIVE MODEL DOES

- **Head removed**
  The part of the simulated domain containing the organizer is discarded.
- **Profiles inherited**
  The remaining fragment keeps the activator and inhibitor profiles present before the cut.
- **Wound response omitted**
  The geometry changes, but no additional biological signal is introduced.

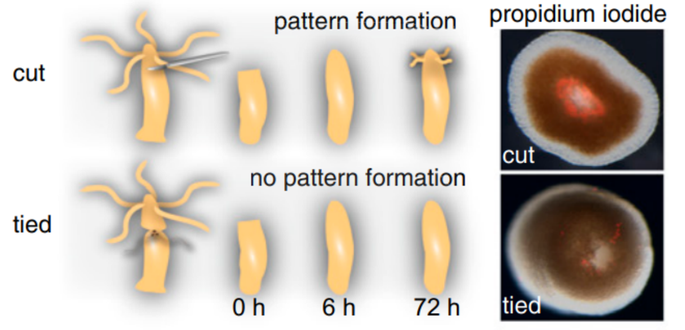
LIVE OUTCOME

THE HEAD DOES NOT REGROW

- **The residual signal fades**
  The inherited headless profile relaxes toward the zero state.
- **No organizer is initiated**
  Nothing pushes the fragment into the basin of a head-forming pattern.

Changing the geometry is not the same as activating a wound response

How the head is removed determines whether it regenerates



Tursch et al., *PNAS* 119, e2204122119 (2022), Fig. 1A.

- **Cut: wound present**
  Cutting disrupts epithelial integrity. Pattern formation follows and a head regenerates.
- **Tied: wound avoided**
  Careful ligation removes the head while preserving epithelial integrity. Regeneration fails.
- **Same loss, different outcome**
  Head removal is not sufficient; tissue injury supplies the initiating signal.

The wound pushes the system out of the zero basin

WITHOUT A WOUND

- **Small morphogen perturbations**
  remain inside the basin of attraction of the zero state.
- **The system relaxes back**
  to the headless state after the perturbation disappears.

AFTER WOUNDING

- **A large local morphogen perturbation**
  is generated near the wound site.
- **The perturbation crosses the basin boundary**
  and moves the system toward the head-organizer attractor.

Wounding provides the large perturbation needed to escape the basin of attraction of the zero state

One cut, two outcomes

Both fragments inherit restricted profiles and receive the same localized wound rule at the new boundary

FRAGMENT WITH THE HEAD

- **Organizer inherited**
  The original peak remains self-maintaining.
- **Wound filtered by context**
  Long-range inhibition prevents the cut end from becoming a second head.

one head remains

HEADLESS FRAGMENT

- **Without the wound input**
  The inherited profile decays toward the zero state.
- **With the wound input**
  The local pulse crosses the threshold and initiates a persistent organizer.

a new head forms

The wound initiates regeneration, the existing pattern determines where it succeeds

Perfect model?

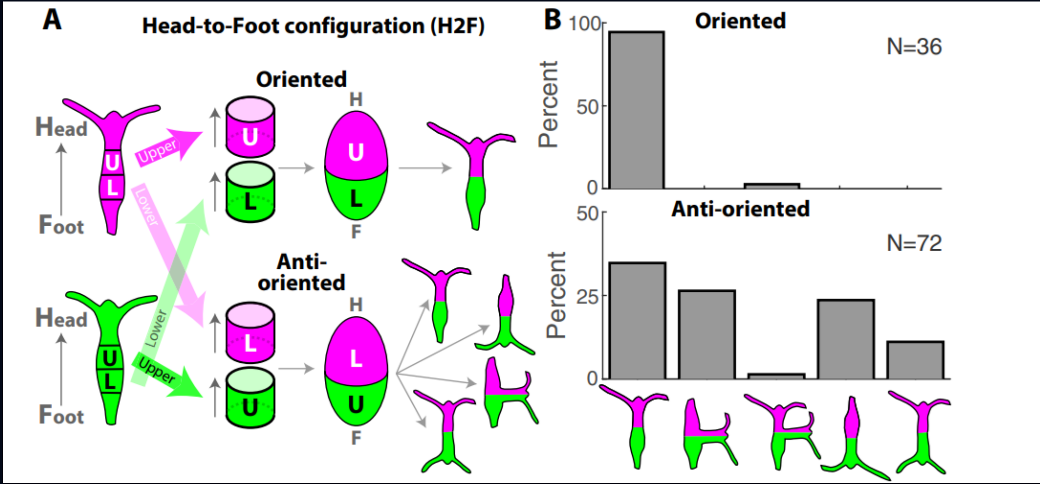

Well yes, but actually no

05 GROWTH, INJURY & SOURCE DENSITY

Source density

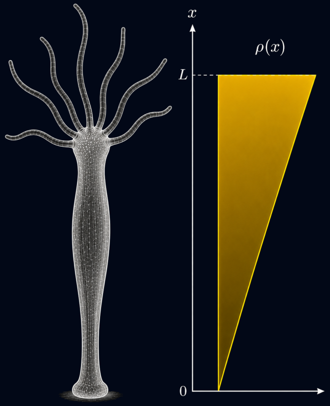
Nothing in the equations distinguishes parts of the tissue

The same two rings, glued in two different orders



Livshits, Garion, Maroudas-Sacks, Shani-Zerbib, Keren & Braun, *Sci. Rep.* **12**:13368 (2022), Fig. 2A–B.

Missing ingredient

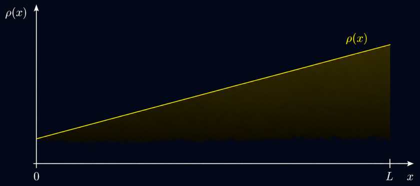


- **How do we tell two pieces of tissue apart?**
  Nothing in the equations distinguishes tissue taken from near the head from tissue taken from near the foot.
- **Positional information**
  Each piece carries a memory of where it sat along the body axis of the parent animal.
- **$\rho(x)$ influences the production of both components**
  The same graded factor multiplies the production of the activator and of the inhibitor.

Model with $\rho(x)$

Let $\Omega = [0, 1] \subset \mathbb{R}$ be an interval

$$\partial_t u = 0.01\Delta u + \rho(x)\,\frac{1.5u^2}{1+v} - 0.5u$$
$$\partial_t v = \Delta v + \rho(x)\,2u^2 - v$$



- **Initial conditions**
  $u(x, 0) = u_0(x)$ and $v(x, 0) = v_0(x)$
- **No-flux boundaries**
  $\partial_x u(x, t) = \partial_x v(x, t) = 0$ at $x = 0$ and $x = 1$

Hard to analyse
A space-dependent coefficient breaks the tools built up so far.

No positive constant steady state
Only the trivial state survives, and every earlier section linearised about a positive one.

Theory exists, but it is heavy
Rigorous results are available and considerably more involved.

But where does $\rho(x)$ come from?

- **It is put in by hand**
  We chose the profile ourselves. Nothing in the model produces it.
- **It never changes**
  A prescribed $\rho(x)$ is fixed for all time, so the tissue cannot rewrite what it knows about its own position.
- **Regeneration has to restore it**
  The animal that grows back must carry a proper gradient again, ready for the next cut. A prescribed profile offers no mechanism for that.

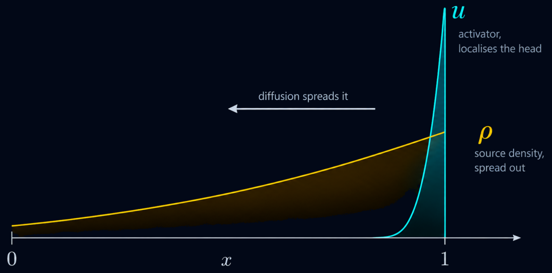
If the source density matters, it has to be a variable of the model

Source density as a variable

$$\partial_t \rho = D_\rho \Delta \rho + u - \rho$$

with $\partial_x \rho = 0$ at $x = 0$ and $x = 1$

- **The activator writes it**
  Wherever the activator is high, the tissue becomes better at producing it.
- **Diffusion spreads it out**
  D_{SD} turns a sharp peak into a profile that varies across the whole domain.



The analysis survives, at a price

$$\partial_t u = 0.01\Delta u + \rho\,\frac{1.5u^2}{1+v} - 0.5u$$
$$\partial_t v = \Delta v + \rho\,2u^2 - v$$
$$\partial_t \rho = D_{SD}\Delta\rho + u - \rho$$

- **More computation**
  Three components instead of two: the characteristic polynomial is cubic and the algebra grows accordingly.
- **Similar criteria**
  Turing instability appears in the same way, and the conditions for it look much as before.

A variable that forgets as fast as it learns

LIVE SIMULATION

Run the three-component system with $\tau = 1$ and watch the source density track the activator instead of outliving it

Making ρ a variable bought us nothing, because it relaxes as quickly as the pattern that writes it

Adding a time scale

$$\tau\, \partial_t \rho = D_\rho \Delta\rho + u - \rho$$

- **Memory**
  Large τ makes the source density outlive the pattern that created it.
- **Separation of scales**
  The activator and inhibitor settle while ρ is still almost frozen.
- **The prescribed profile returns**
  As τ grows, the model approaches the fixed $\rho(x)$ we started with, now as a limit rather than an assumption.

τ does not decide whether a pattern forms, but how long the tissue remembers where it was

Model with source density

LIVE SIMULATION

Run the same system with a slow source density and watch the gradient be inherited, rewritten, and set the site of the new head

The gradient is no longer assumed, the tissue writes it, keeps it, and hands it on

Perfect model?

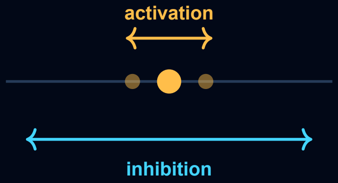

Well yes, but actually no

Recall the Gierer–Meinhardt mechanism

The chemical mechanism uses two interacting substances

$$\partial_t u = D_u \Delta u + \frac{a u^2}{K + v} - \mu_u u$$
$$\partial_t v = D_v \Delta v + b u^2 - \mu_v v$$

- **Local activation**
  The activator promotes its own production over a short range
- **Long-range inhibition**
  A second substance suppresses activation over a larger spatial scale



two spatial scales are encoded
by two diffusible substances

The activator has candidates but the inhibitor remains unclear

ACTIVATOR CANDIDATE

Wnt3

Stable localized Wnt3 expression marks the emerging head organizer

MISSING COMPONENT

?

No biochemical species has been established as the required fast long-range inhibitor

The mathematical inhibitor has a clear role but its biological identity is unresolved

LECTURE MAP

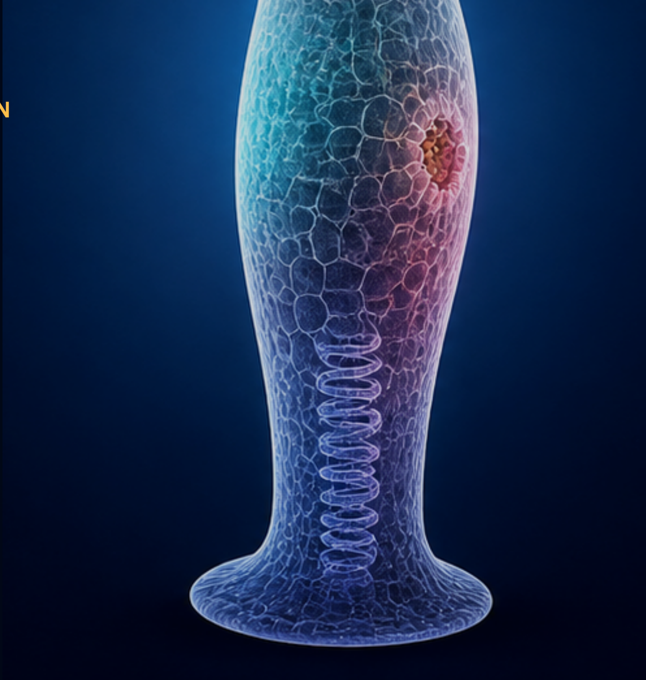
The road from Hydra to pattern formation

From biological observations to mathematical mechanisms.

06 MECHANOCHEMICAL PATTERN FORMATION

Why mechanics?

The classical inhibitor is not identified and mechanics offers an alternative long-range coupling



The activator has candidates but the inhibitor remains unclear

ACTIVATOR CANDIDATE

Wnt3

Stable localized Wnt3 expression marks the emerging head organizer

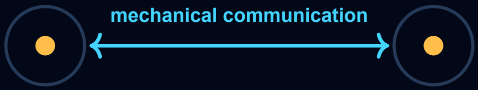
MISSING COMPONENT

?

No biochemical species has been established as the required fast long-range inhibitor

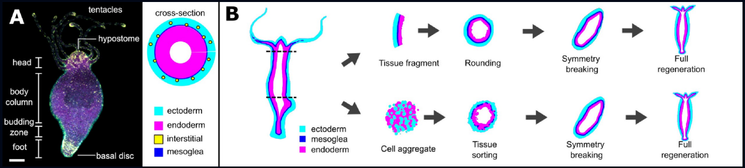
Long-range inhibition need not be chemical

- **Chemical transport**
  A molecule would have to transmit inhibition across the regenerating tissue
- **Mechanical coupling**
  Stress and deformation can couple distant parts of a closed tissue without a second inhibitor
- **Modeling question**
  Can one morphogen and tissue mechanics reproduce local activation and long-range inhibition?



a global constraint can transmit
local changes throughout the tissue

Regenerating spheroid



Wang, Bialas, Goel & Collins, *Integrative and Comparative Biology* 63 (2023), Fig. 1B.

- **Initially symmetric**
  A tissue fragment rounds into a hollow spheroid
- **Symmetry is broken**
  One region becomes the future head organizer
- **A complete axis emerges**
  The localized organizer directs whole-body regeneration

The central question

Can a local morphogen change the mechanics of the entire spheroid?

Can mechanics replace the diffusible inhibitor?

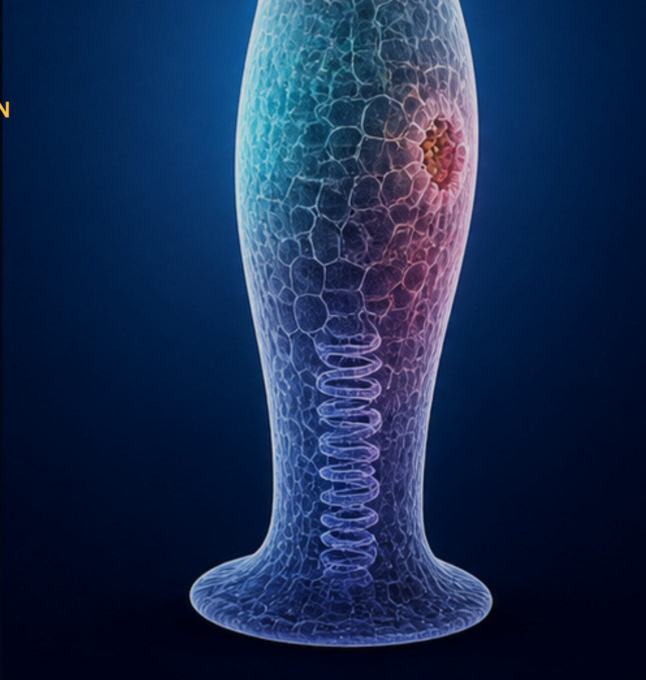
$\Downarrow$

Build the simplest mechanochemical model that can answer this question

06 MECHANOCHEMICAL PATTERN FORMATION

Mechanics during regeneration

Swelling and rupture expose
a two-way coupling between
morphogen signaling and mechanics



One oscillation is a swelling and rupture cycle

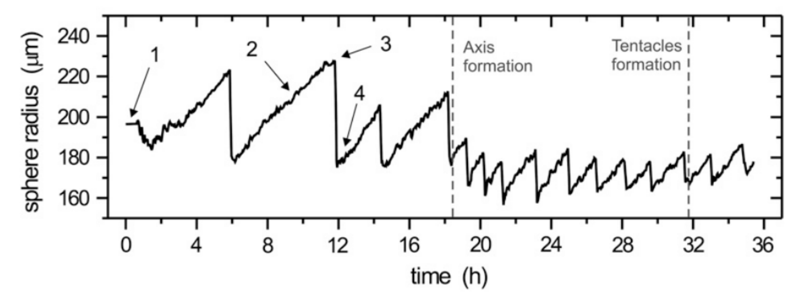
water enters → tissue stretches → tissue ruptures

⇓

the tissue deflates, heals, and the cycle begins again

Osmotic pressure repeatedly converts fluid transport into tissue stretching

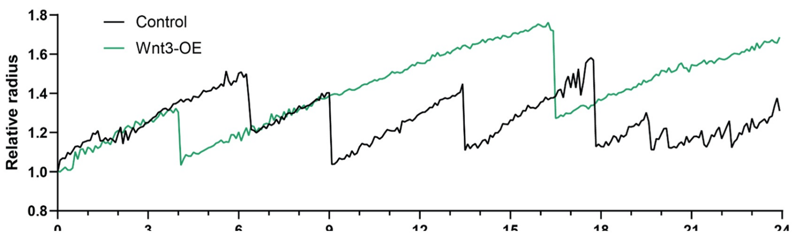
The spheroid radius records repeated rupture events



- **Slow increase**
  Water influx gradually increases the spheroid radius
- **Rapid decrease**
  Rupture releases fluid and abruptly reduces the radius
- **Changing dynamics**
  The oscillation changes as the body axis and tentacles form

Kuecken et al., Biophysical Journal, 2008

Wnt3 changes the oscillatory dynamics



- **Control**
  Repeated rupture limits the gradual increase in radius
- **Wnt3 overexpression**
  The timing and amplitude of rupture events are altered

Morphogen signaling and tissue-scale mechanics cannot be treated independently

Experiments suggest a mechanochemical coupling

TISSUE STRETCHING

⇓ **Wnt3 production**

Stretching correlates with organizer formation and Wnt3 expression

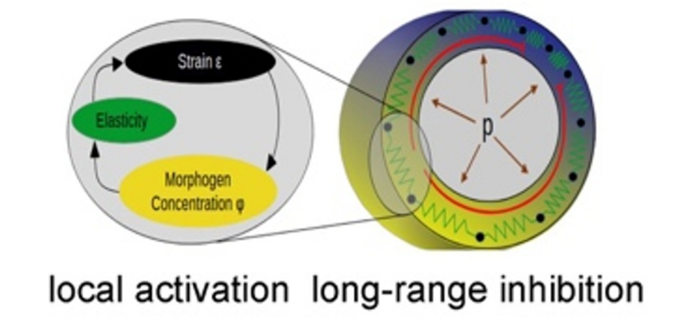
MORPHOGEN

⇓ **Tissue elasticity**

Morphogen signaling changes how strongly the tissue resists deformation

Each side of the feedback modifies the other

Mechanochemical feedback

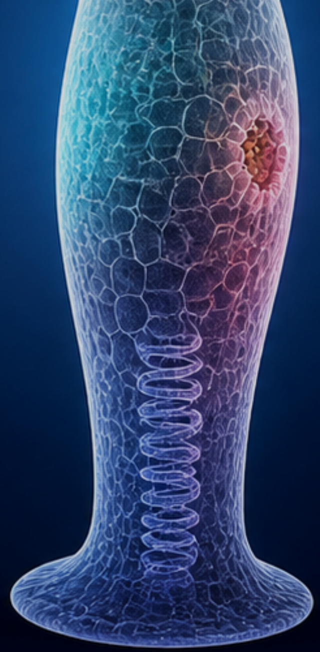


- **Local activation**
  Morphogen changes elasticity and concentrates strain locally
- **Long-range inhibition**
  The closed pressurized tissue shares a finite total deformation
- **Minimal ingredients**
  Morphogen dynamics, mechanics and one global constraint

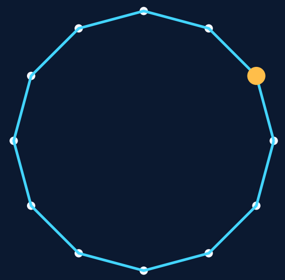
06 MECHANOCHEMICAL PATTERN FORMATION

Elastic spring model

A closed ring of springs reveals
how local elasticity produces
a global redistribution of strain

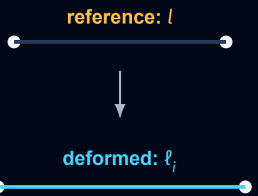


Local changes redistribute strain around the ring



- **Move one node**
  A local displacement changes several neighboring springs
- **Change one stiffness**
  A softer spring takes a larger share of the deformation
- **Close the ring**
  The total extension cannot be assigned independently at every point

The energy of a deformed spring



Spring i may stretch or compress relative to its reference length

Hooke's law assigns the spring energy

$$\boxed{E_i = \frac{1}{2} E(u_i) \left(\ell_i - l\right)^2}$$

- l
  Reference length of the spring, before deformation
- ℓ_i
  Deformed length of spring i, after stretching or compression
- $E(u_i)$
  Elastic modulus set by the local morphogen concentration u_i

High morphogen concentration increases the spring's elasticity

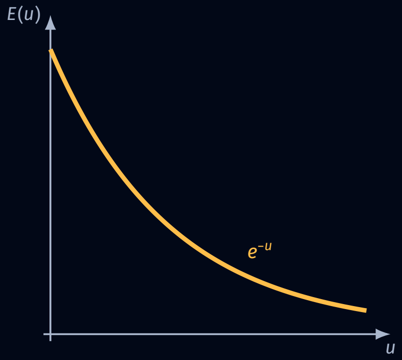
Morphogen concentration influences local stiffness

Experimental measurements support this trend: tissue softens as morphogen concentration rises

$$E'(u) < 0$$

- **High morphogen**
  The elastic modulus is smaller and the tissue is softer
- **Low morphogen**
  The elastic modulus is larger and the tissue is stiffer

$$E(u) = e^{-u}$$



The local feedback is positive

higher u $\Rightarrow$ **smaller** $E(u)$ $\Rightarrow$ **larger strain** $\Rightarrow$ **more** u

- **Local activation**
  A small local increase in morphogen makes the same region easier to stretch
- **Missing ingredient**
  We still need to determine how the strain is distributed around the entire ring

The total discrete energy is a sum over all springs

Every spring contributes its local elastic energy

$$E_N = \sum_{i=0}^{N-1} E_i = \sum_{i=0}^{N-1} \frac{1}{2} e^{-u_i} \left(\ell_i - l\right)^2$$

- **Local contribution**
  Each term depends only on the deformed length and morphogen at spring *i*
- **Global configuration**
  The total energy depends on how deformation is shared by all springs

Inflation and energy minimization

- **Local stiffness**
  The stiffness of each spring is set by the local morphogen concentration
- **Osmotic driving force**
  Tension is induced as the spheroid stretches while inflating with pumped-in water
- **Heterogeneous response**
  Springs stretch by different amounts according to their individual elasticity
- **Energy minimization**
  Springs stretch so as to minimize the total energy of the ring while it keeps its circular shape

Passing with the number of springs to the limit

Passing with the strain

$$\varepsilon_i = \frac{\ell_i - l}{l} \quad \Rightarrow \quad \varepsilon(x) = \lim_{l \to 0} \frac{\ell_i - l}{l}$$

Passing with the energy

$$E_N = \sum_{i=0}^{N-1} \frac{1}{2} e^{-u_i} (\ell_i - l)^2 \quad \Rightarrow \quad E(x) = \frac{1}{2} \int_0^L e^{-u(x)} \varepsilon(x)^2 \, dx$$

Passing with the constraint

$$\ell - L = \sum_{i=0}^{N-1} \varepsilon_i \, l \quad \Rightarrow \quad \ell - L = \int_0^L \varepsilon(x) \, dx$$

The tissue minimizes the energy instantly

- **Instantaneous minimization**
  Mechanics relaxes much faster than the morphogen dynamics, so for every morphogen profile the ring is already at the minimum of its elastic energy

- **Any other configuration costs more**
  Every other way of distributing the stretching along the ring gives a strictly larger total energy

- **Vanishing variation**
  Mathematically this requires the derivative of the energy with respect to the strain to vanish

In one dimension the condition reduces to

$$\boxed{\partial_x\left(e^{-u}\varepsilon\right) = 0}$$

Morphogen dynamics

$$\boxed{\partial_t u = D\,\partial_{xx} u - u + \kappa\varepsilon}$$

Boundary conditions

$$u(0,t) = u(L,t)$$
$$\partial_x u(0,t) = \partial_x u(L,t)$$

Initial condition

$$u(x,0) = u_0(x)$$

- **Diffusion** $D\,\partial_{xx} u$
  The morphogen spreads along the ring, with diffusion coefficient D
- **Degradation** $-u$
  The morphogen decays at a constant rate, normalized to one
- **Mechanical production** $\kappa\varepsilon$
  Stretched tissue produces more morphogen, with mechanosensitivity κ

The full model

Let Ω be a circle of length one

$$\partial_t u = D\partial_{xx} u - u + \kappa\varepsilon$$
$$0 = \partial_x(e^{-u}\varepsilon)$$
$$\ell - L = \int_0^L \varepsilon(x,t)\,dx$$

supplemented with periodic boundary conditions and an initial morphogen profile

- **One evolution equation**
  Only the morphogen has its own time dynamics
- **Instantaneous mechanics**
  Strain is determined by equilibrium at every time
- **One global constraint**
  The prescribed total extension couples the entire ring

Mechanical equilibrium determines the strain explicitly

$$\partial_x\left(e^{-u(x,t)}\varepsilon(x,t)\right) = 0 \quad + \quad \ell - L = \int_0^L \varepsilon(x,t)\, dx$$

$$\Downarrow$$

$$\boxed{\varepsilon(x,t) = \frac{(\ell - L)\, e^{u(x,t)}}{\displaystyle\int_0^L e^{u(y,t)}\, dy}}$$

One nonlocal equation for the morphogen

Let Ω be a circle of length one

$$\boxed{\partial_t u = D\partial_{xx} u - u + \kappa \frac{e^{u(x,t)}}{\int_\Omega e^{u(y,t)}\, dy}}$$

Boundary conditions

$$u(0,t) = u(1,t)$$
$$\partial_x u(0,t) = \partial_x u(1,t)$$

Initial condition

$$u(x,0) = u_0(x)$$

- **Single equation**
  The whole mechanochemical model reduces to one equation for the morphogen
- **An unusual situation**
  A single reaction–diffusion equation cannot normally produce patterns
- **Nonlocal term**
  The nonlocal coupling plays the crucial role and replaces the missing inhibitor

The special structure of this equation makes it far more tractable analytically than the Gierer–Meinhardt model

One equation realizes the activator–inhibitor paradigm

LOCAL ACTIVATION

$$e^{u(x,t)}$$

A region with more morphogen becomes softer and receives more strain-driven production

LONG-RANGE INHIBITION

$$\int_0^1 e^{u(y,t)}\,dy$$

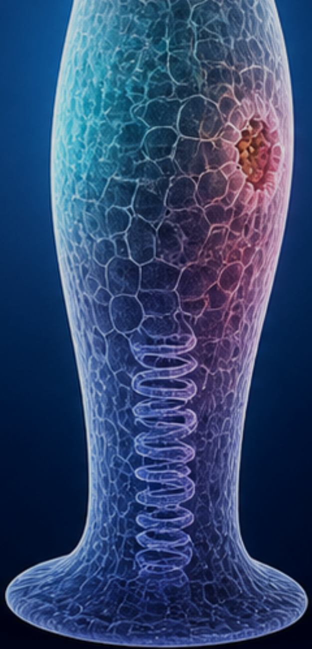
Increased production in one region reduces the share of extension available elsewhere

Mechanical equilibrium converts a local positive feedback into global competition

06 MECHANOCHEMICAL PATTERN FORMATION

Steady states and stability

Variational structure determines when patterns exist and which stationary profiles can be stable



Stationary solutions

$$0 = D\partial_{xx}U - U + \kappa \frac{e^{U(x)}}{\int_0^1 e^{U(y)}\,dy}$$

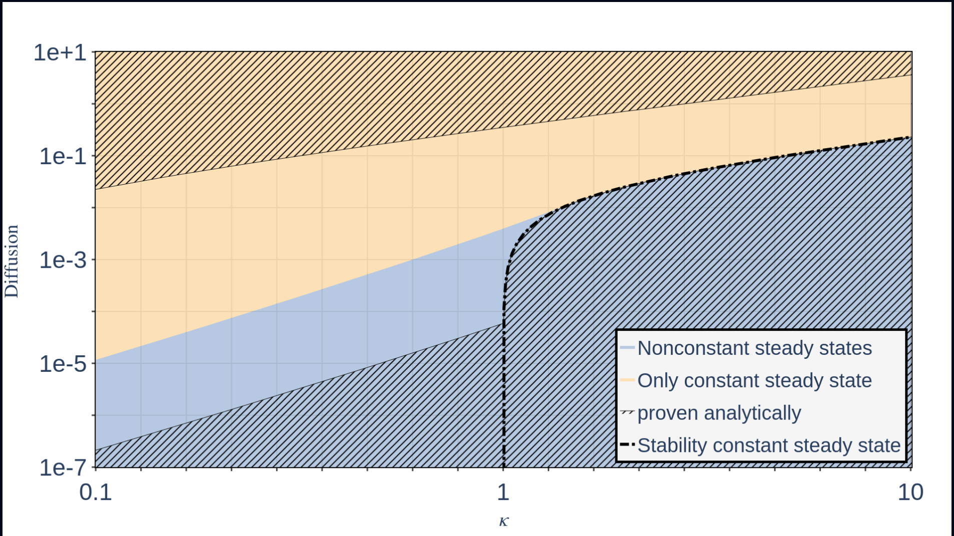
- **Homogeneous solution**
  $\bar{U} \equiv \kappa$ is always a stationary solution
- **Patterned solution**
  A nonconstant periodic function $U(x)$ represents a spatial pattern

THEOREM

For every $\kappa > 0$ there exist constants $0 < D_{\min}(\kappa) < D_{\max}(\kappa)$ such that

- $0 < D < D_{\min}(\kappa)$
  A nonconstant stationary solution exists and is Lyapunov stable in L^2
- $D > D_{\max}(\kappa)$
  $\bar{U} \equiv \kappa$ is the unique stationary solution and is globally asymptotically stable

The parameter space



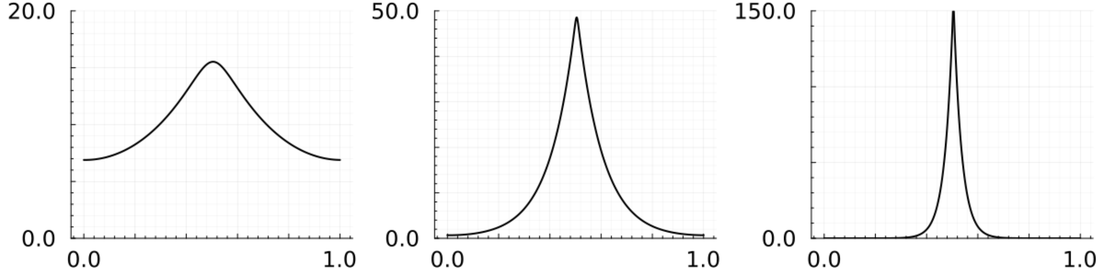
Stable mechanochemical patterns are necessarily unimodal

$$0 = D\partial_{xx}U - U + \kappa \frac{e^{U(x)}}{\int_0^1 e^{U(y)}\, dy}$$

THEOREM

Every nonconstant stationary solution with two or more peaks is unstable. A stable stationary solution can therefore have at most one peak.

Diffusion changes the width but not the number of peaks

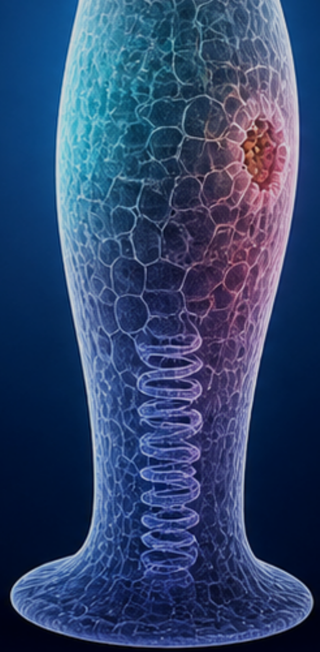


- **Smaller diffusion**
  The stable peak becomes sharper and more localized
- **Larger diffusion**
  The stable profile becomes broader
- **Robust selection**
  The stable pattern retains a single dominant peak

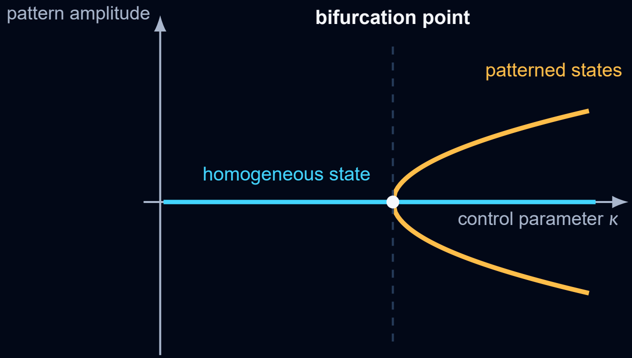
06 MECHANOCHEMICAL PATTERN FORMATION

Bifurcations and oscillations

Stationary branches emerge when a homogeneous state changes stability and can generate tissue oscillations



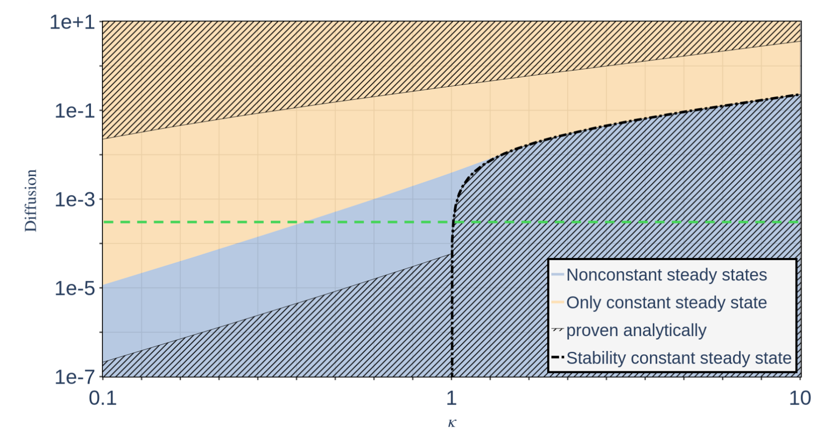
A bifurcation marks a qualitative change in stationary states



- **Before the threshold**
  Only the homogeneous stationary state is visible nearby
- **At the threshold**
  A spatial mode becomes neutral and a patterned branch can emerge
- **After the threshold**
  The number or stability of stationary states has changed

Bifurcation diagram

$$\partial_t u = D\partial_{xx} u - u + \kappa \frac{e^u}{\int_0^1 e^u}$$



Biffuration types

The constant steady state exhibits a pitchfork bifurcation at every

$$\kappa_n = 1 + 4\pi^2 n^2 D \qquad \text{for } n \in \mathbb{N}$$

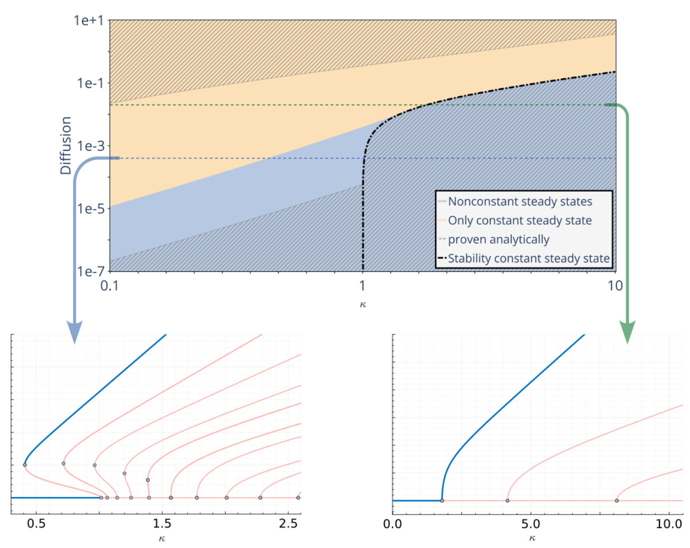
In particular the constant steady state $\bar{U} \equiv \kappa$ loses stability at $\kappa_1 = 1 + 4\pi^2 D$

THEOREM

For fixed diffusion $D > 0$ there exists a branch of stationary solutions $U(s)$ with $\kappa = \kappa(s)$ and $s \in (-\delta, \delta)$ satisfying $\kappa(0) = \kappa_n$ and $\kappa'(0) = 0$. Moreover, if

- $\kappa_n > 3/2$ then $\kappa''(0) > 0$ (supercritical pitchfork bifurcation)
- $\kappa_n < 3/2$ then $\kappa''(0) < 0$ (subcritical pitchfork bifurcation)

Diffusion determines the type of each bifurcation



- **Small diffusion**
  The branch can bend toward smaller values of κ
- **Larger diffusion**
  The branch can emerge toward larger values of κ

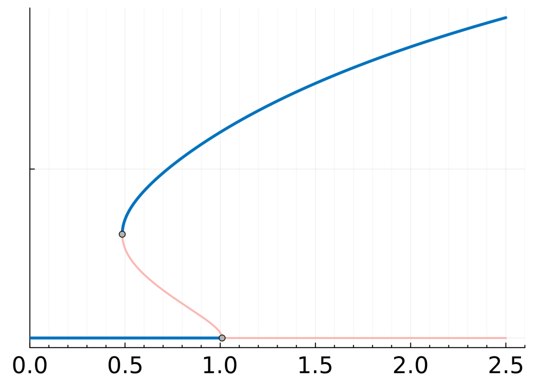
Tissue length turns the stationary picture into dynamics

Let $\ell(t)$ denote the current circumference and L its reference value

$$\boxed{\partial_t u = D\partial_{xx} u - u + \kappa\big(\ell(t) - L\big)\frac{e^{u(x,t)}}{\int_0^1 e^{u(y,t)}\,dy}}$$

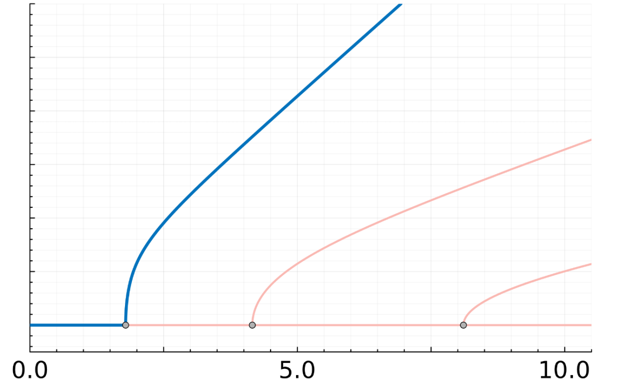
- **Morphogen rises**
  The tissue softens and the mechanical equilibrium shifts
- **The spheroid expands**
  Increased strain strengthens morphogen production
- **Feedback relaxes**
  Degradation and mechanical restoration can reverse the expansion

Subcritical feedback



- **Coexisting states**
  A homogeneous and a patterned state may be available for the same parameter value
- **Sudden transition**
  Crossing a fold can force the system to jump to a distant branch

Supercritical feedback



- **Continuous onset**
  The patterned amplitude grows continuously from the homogeneous state
- **Smooth mechanical response**
  Tissue deformation can track the changing stationary branch

Mechanics can realize an activator–inhibitor mechanism without a chemical inhibitor

- **Local positive feedback**
  Stretch increases morphogen production and morphogen softens the tissue
- **Global mechanical competition**
  The fixed total length couples every point through one nonlocal constraint
- **Single organizer**
  Stable nonconstant stationary states contain one dominant peak

$$\text{morphogen} \Longrightarrow \text{softening} \Longrightarrow \text{strain} \Longrightarrow \text{morphogen}$$

A single morphogen equation coupled to mechanics can implement local activation and long-range inhibition

Perfect model?


Well yes, but actually no

Many desirable properties, and one hard limit

- **The model does almost everything we wanted**
  A single morphogen coupled to tissue mechanics delivers local activation and long-range inhibition, without invoking a second chemical that nobody has ever measured.
- **But it always produces exactly one peak**
  Every nonconstant stationary solution with two or more peaks is unstable. This is a theorem, not a matter of choosing parameters differently.
- **The experiments say otherwise**
  Strongly stretched spheroids are observed to carry more than one peak, which the model cannot reproduce for any choice of parameters.

Story still continues …

Thank you

for your attention